BASEBOARD

DIGITAL SHARK TECHNOLOGY		
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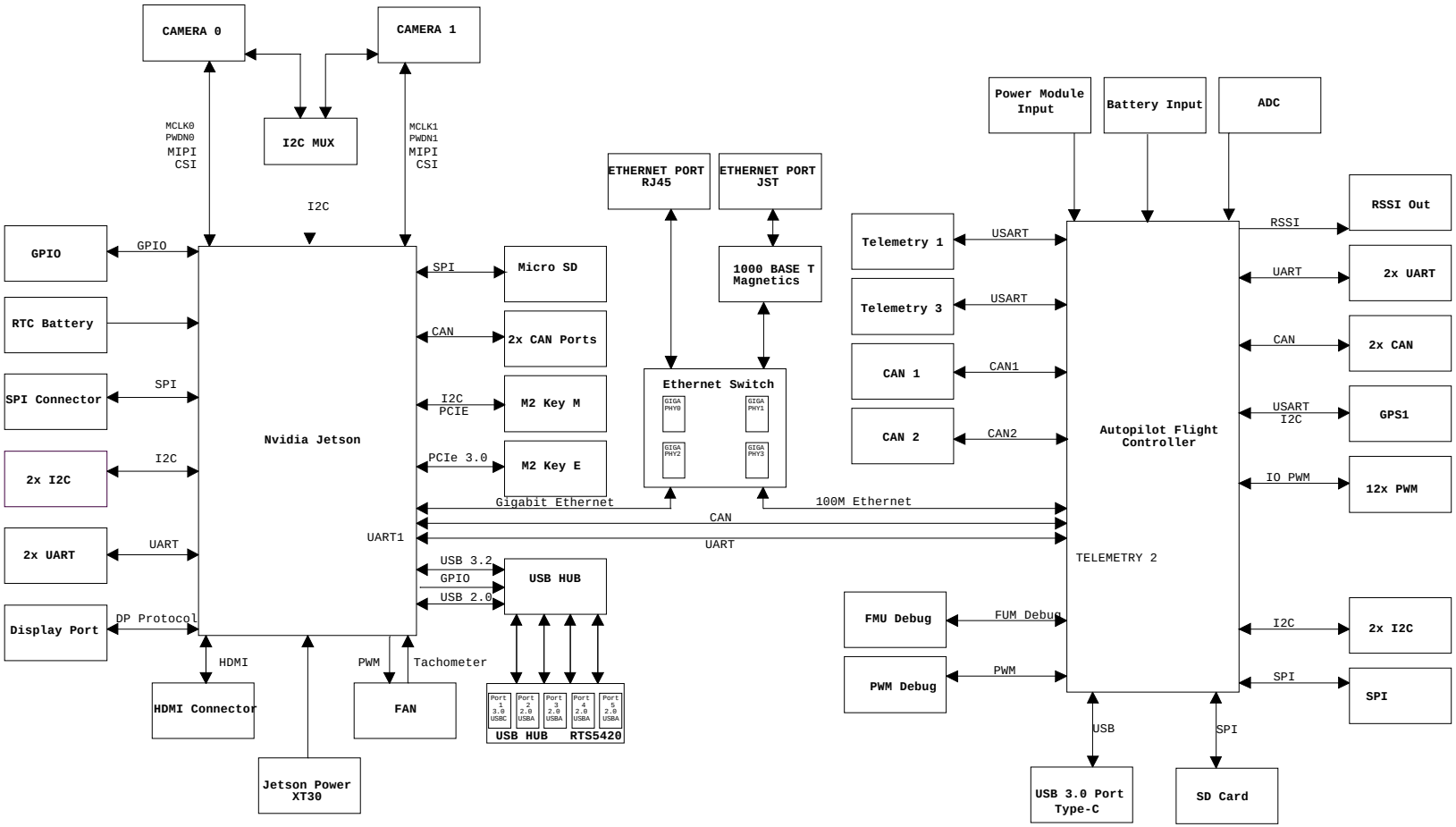
Revision History

Revision	Date	By	Description
Draft	16 AUG 2024	Aakash S J	Draft version
V1.0	11 SEPT 2024	Aakash S J	Updated with changes after review
V1.1	20 MAR 2025	Aakash S J	Updated with changes after review and reconfiguration of MCU
V1.2	28 AUG 2025	Aakash S J	Updated with changes after client requests. Updated BOM for better suitablity

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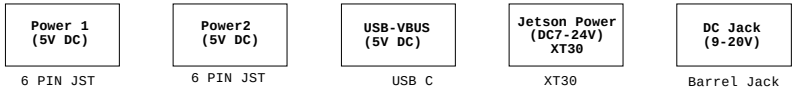
BLOCK DIAGRAM



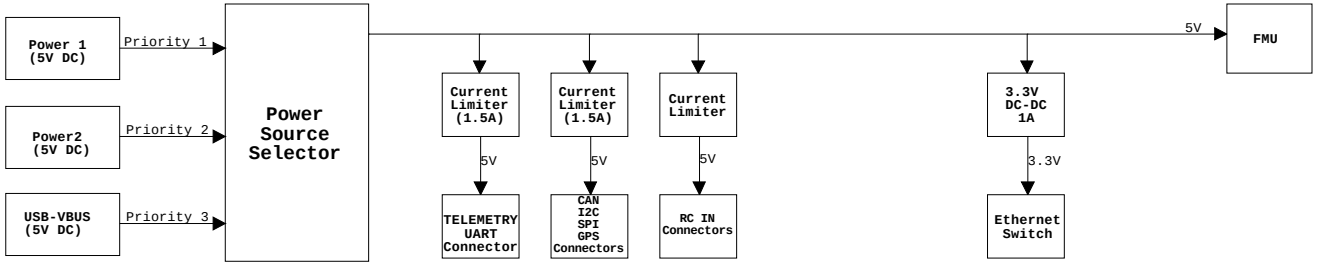
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POWER TREE

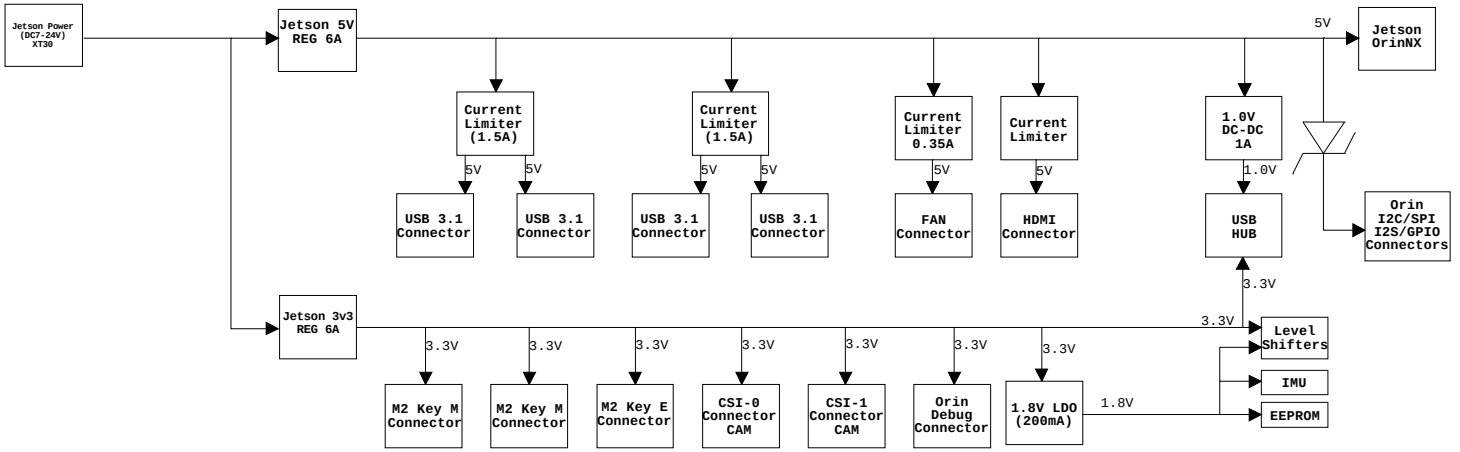
POWER INPUTS



FMU POWER SUPPLY



JETSON POWER SUPPLY

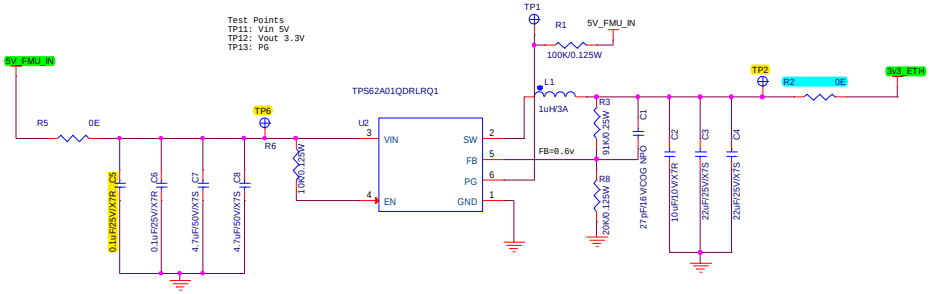


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FCU POWER REGULATORS

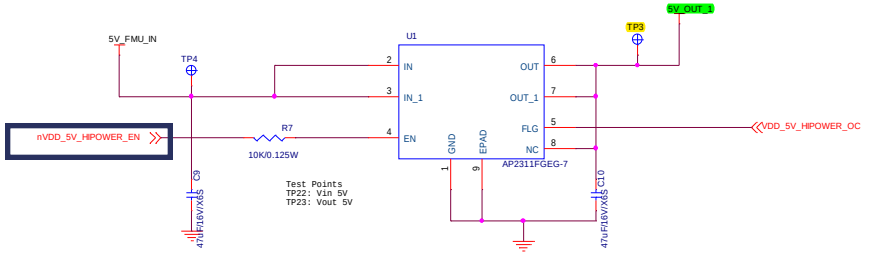
3.3V ETH POWER SUPPLY

Remove R2- during Board bring-up



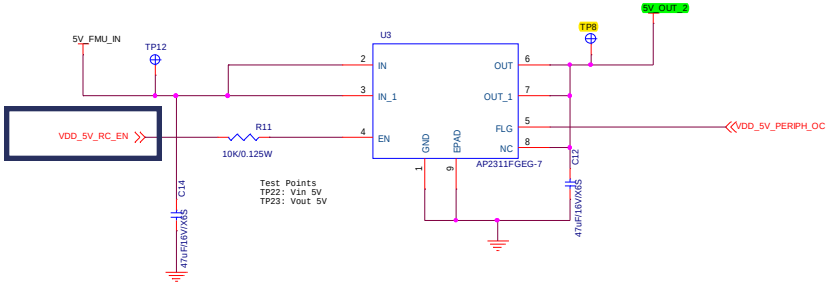
5V Current Limiter 1

2A Limit, UART,TELEM



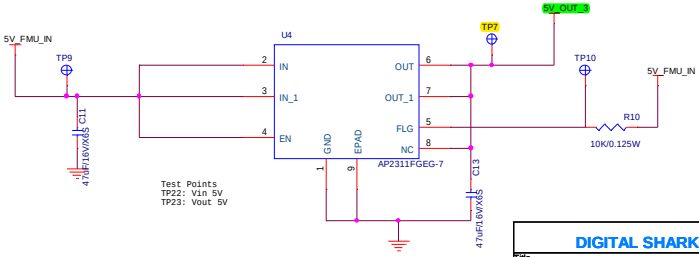
5V Current Limiter 2

2A Limit, I2C, CAN,SPI, GPS



5V Current Limiter 3

2A Limit, RC IN

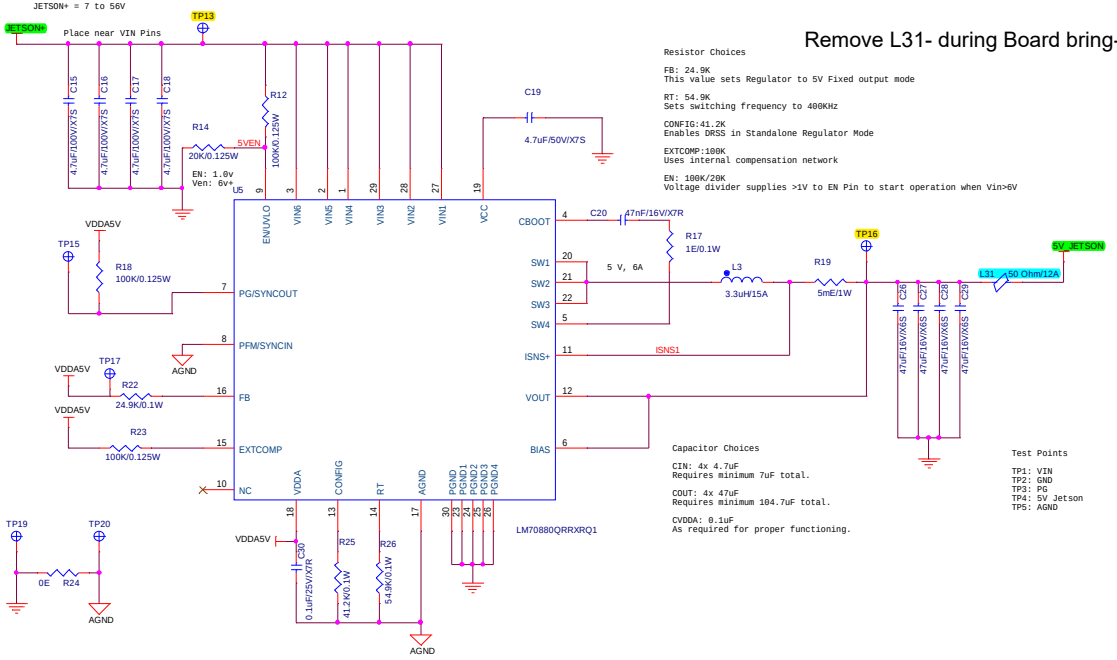


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JETSON POWER REGULATORS

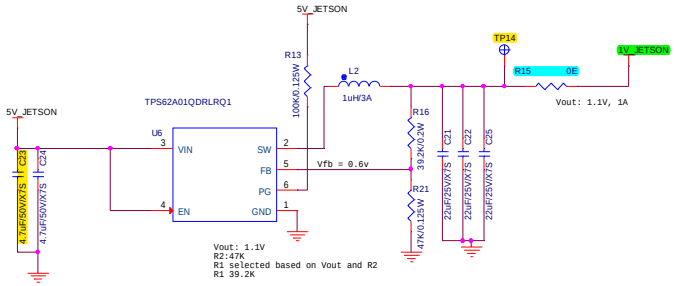
5V JETSON POWER SUPPLY



Remove L31- during Board bring-up

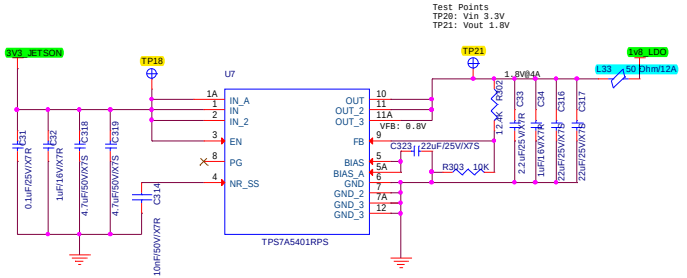
1V JETSON POWER SUPPLY

Remove R15- during Board bring-up

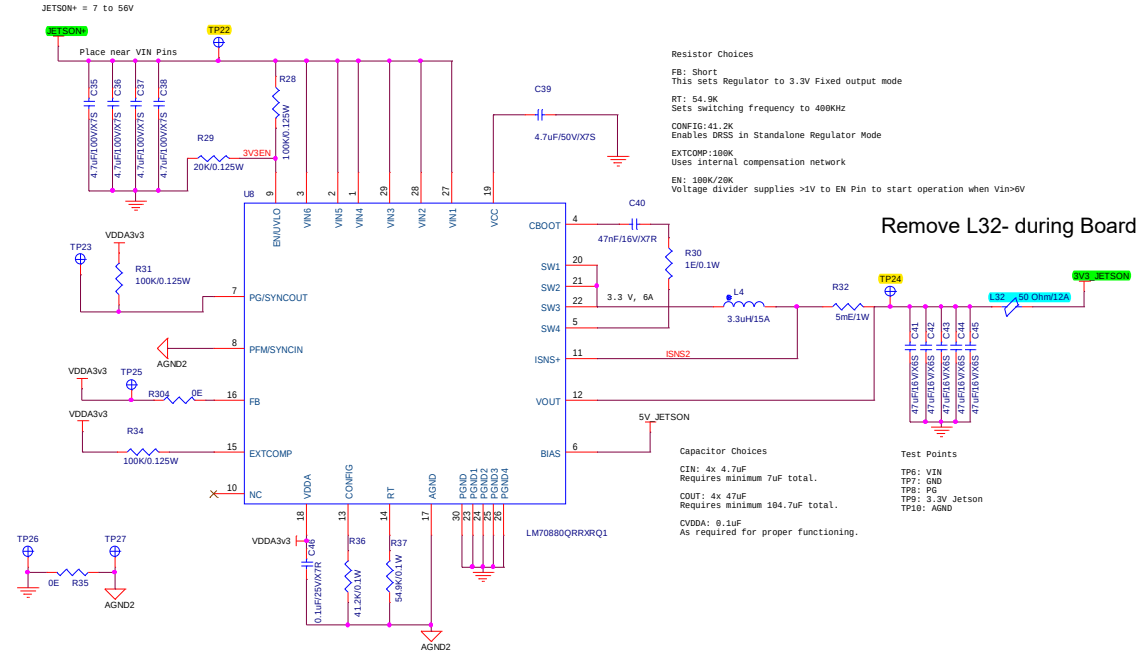


Remove L33- during Board bring-up

1.8V LDO



3.3V JETSON POWER SUPPLY



Remove L32- during Board bring-up

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Schematic Error:
In place of V1,V2 & V3 -
the following Net names to be replaced.
V1 = 5V_POWERA_RVP
V2 = 5V_POWERC_RVP
V3 = FMU_VBUS

VOLTAGE PRIORITY SELECTOR

TP28 = 1.08 V
TP31 = 0.95 V

UV threshold = 4.64V
OV threshold= 5.24V

PROTECTION CIRCUIT

Decoupling Capacitors

Output Capacitors

Input Capacitors

ESD Protection

Total Resistance:

$$R_{total} = 309k + 10k + 75k = 394k\Omega$$

Voltage at TP31

TP31 is between R39 and R40, so:

$$R_{below} = 75k\Omega$$

$$V_{TP31} = V1 \times \frac{75}{394} \approx 0.19 \times V1$$

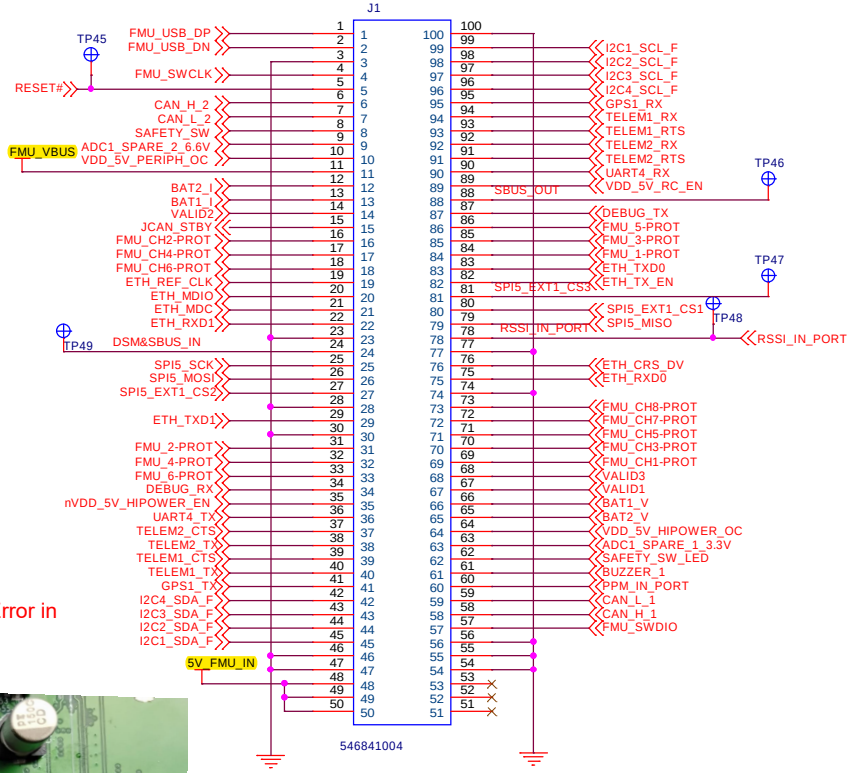
If V1 = 5V:

$$V_{TP31} \approx 0.95V$$

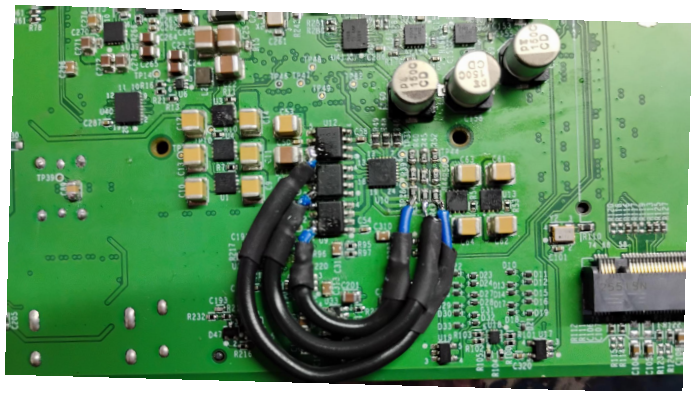
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100P CONNECTOR



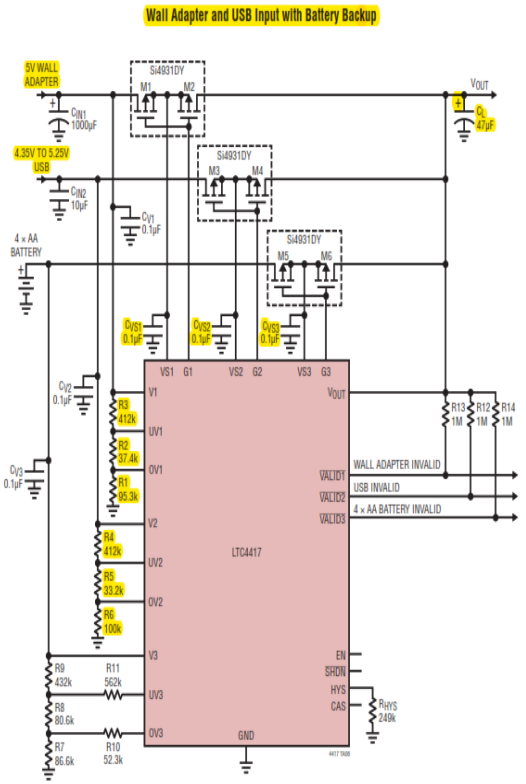
Wire Jumpers Soldered to Compansate Schematic Error in "Voltage Priority Selector"



Resistor Values for "Voltage Priority Selector"

LTC4417

TYPICAL APPLICATIONS

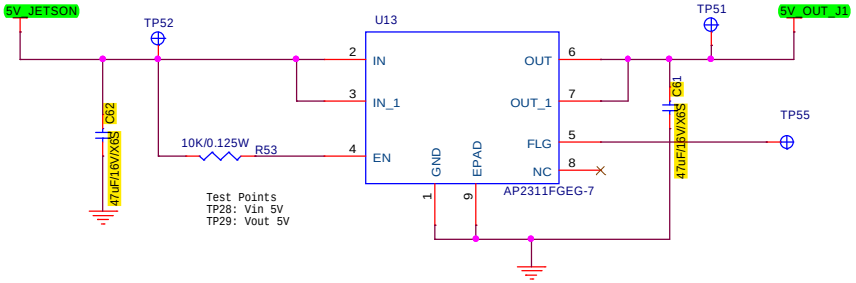


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JETSON CURRENT LIMITERS

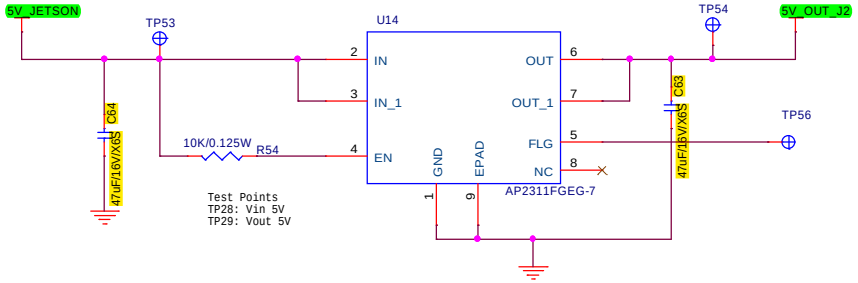
5V Current Limiter Jetson 1

2A Limit, USB Connectors A 1



5V Current Limiter Jetson 2

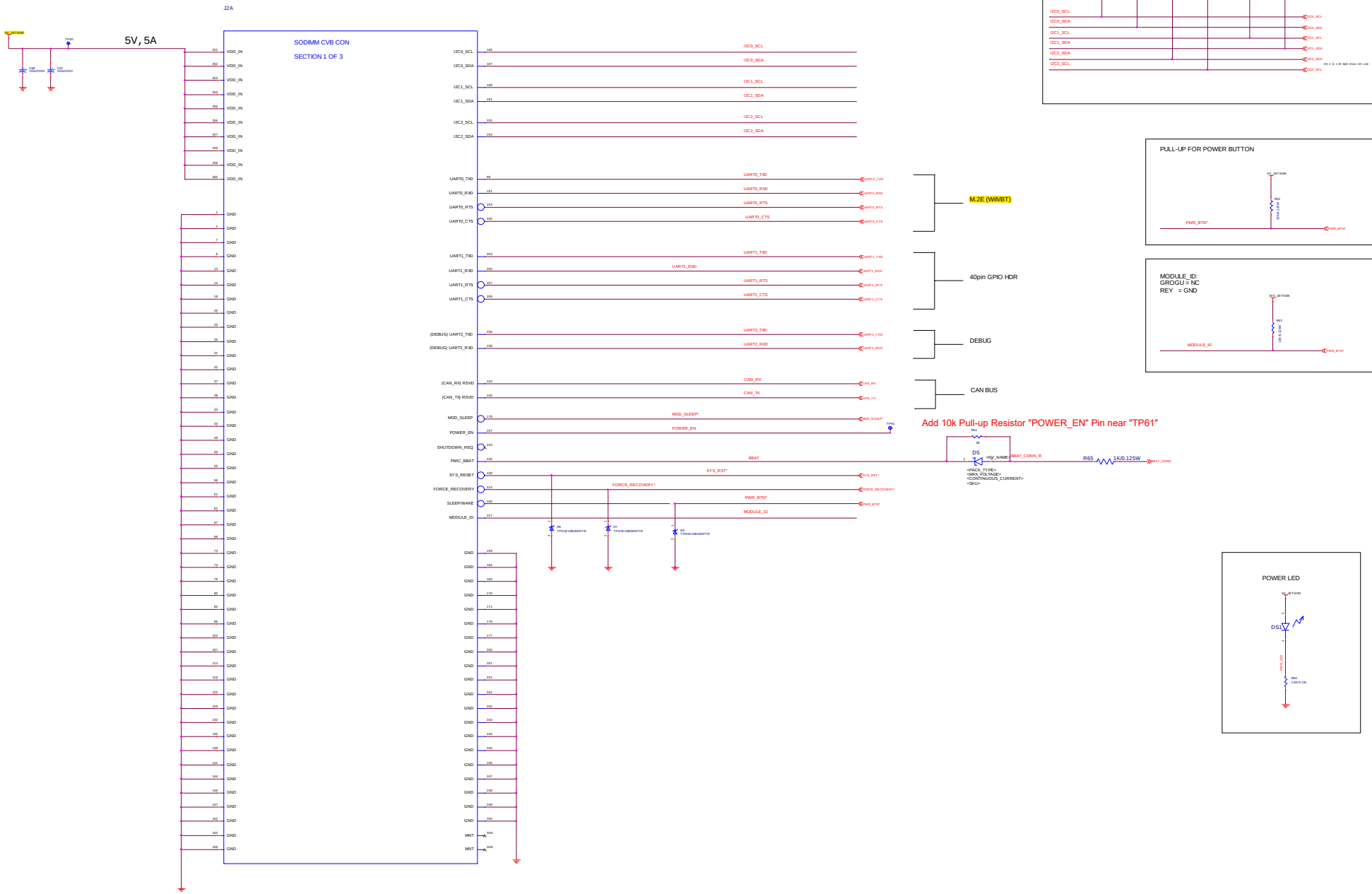
2A Limit, USB Connectors A 2



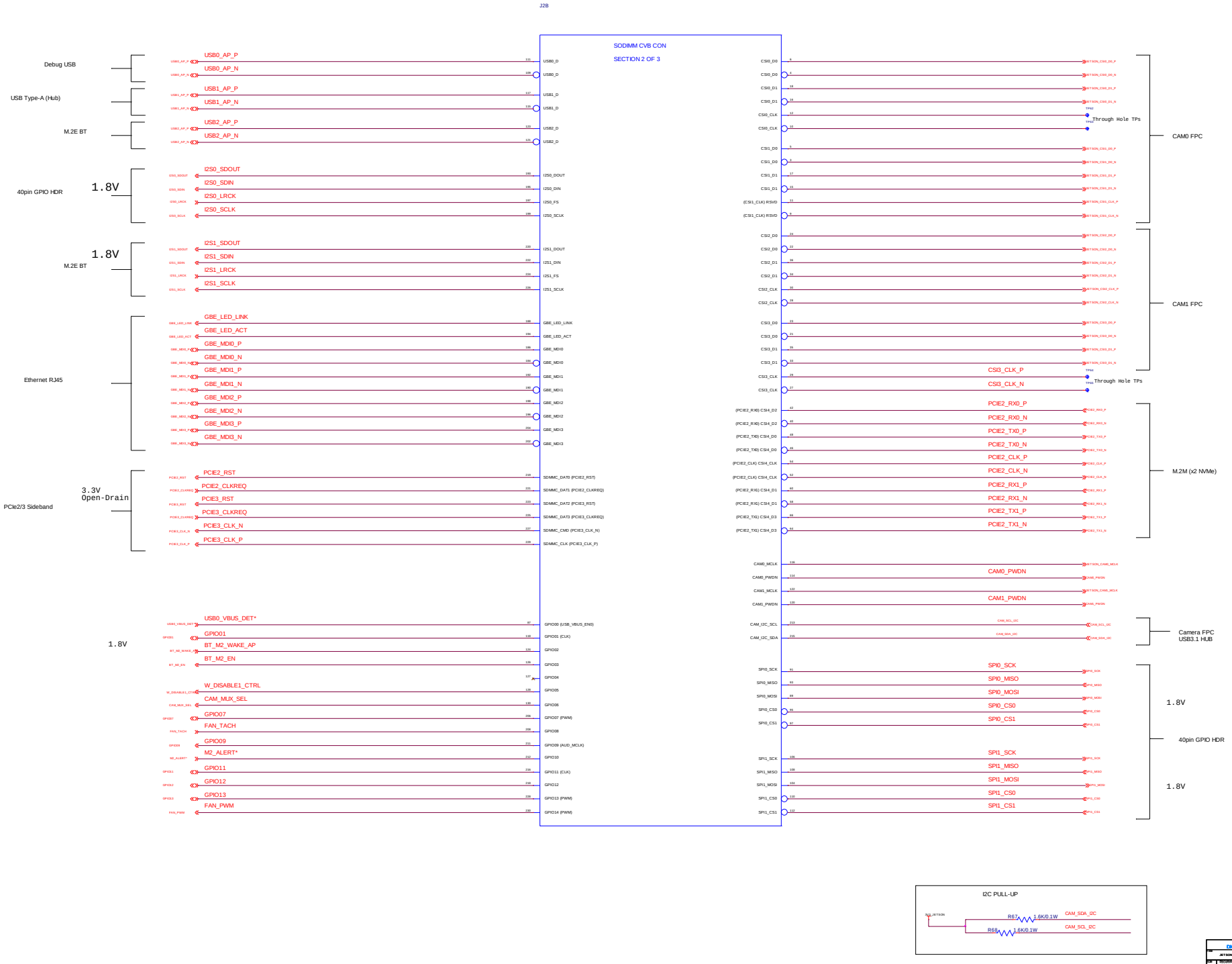
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SODIMM CONNECTOR 1/3

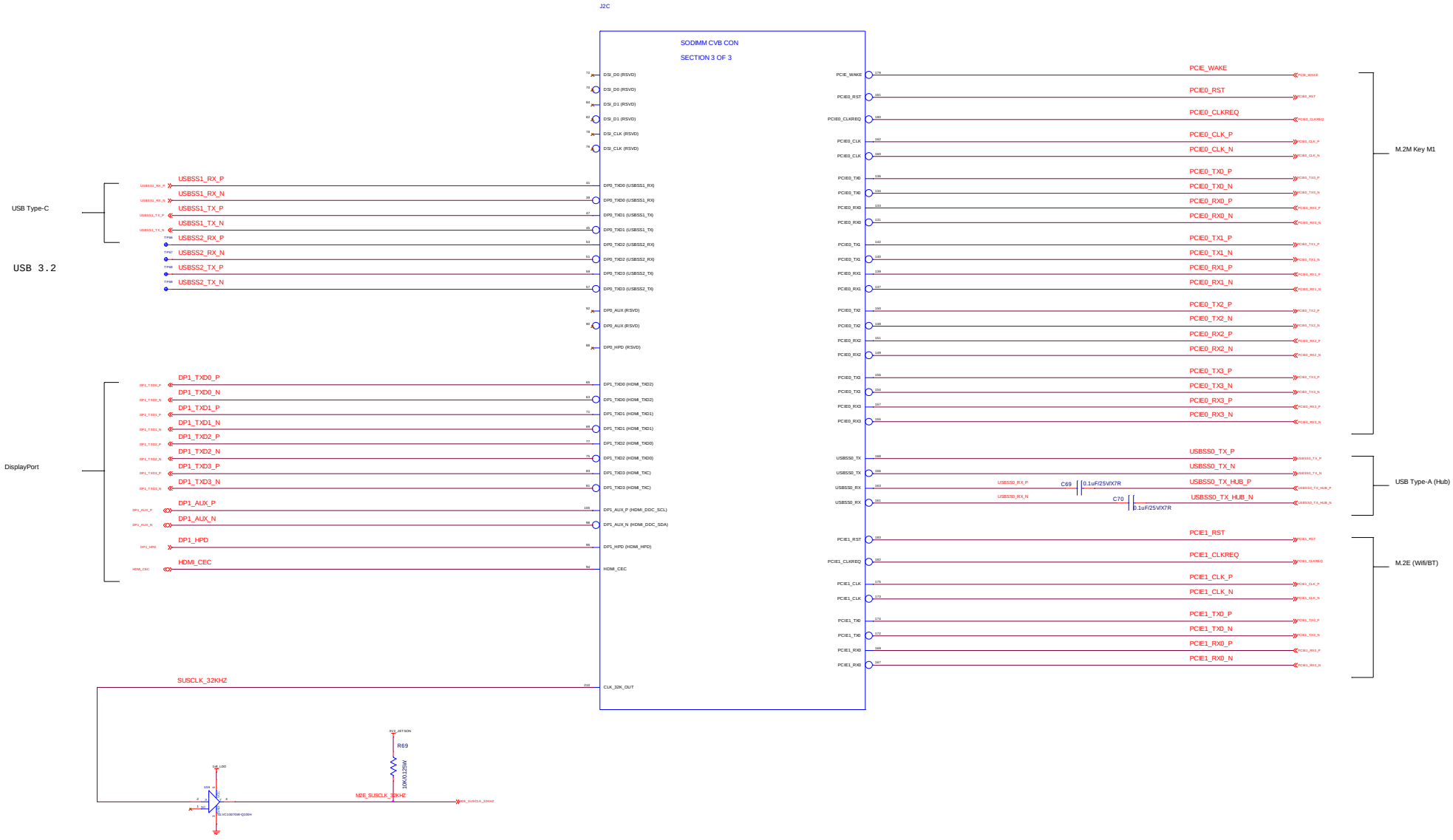
UART, I2C, CAN and GPIOs



GPIOs, USB2.0, I2S, SPI, GBe and CSI



DP, USBSS and PEX

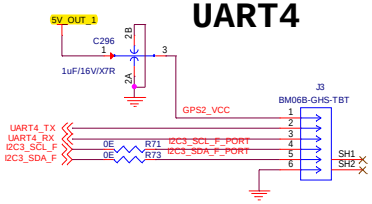


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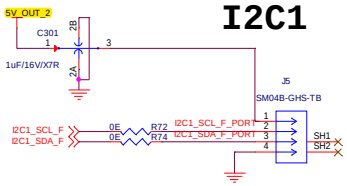
DR	JETSON SODIMM 3	
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FCU I/O Connectors

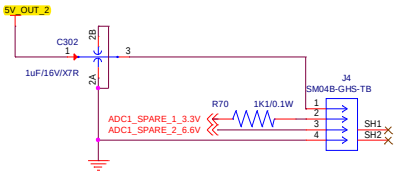
UART4



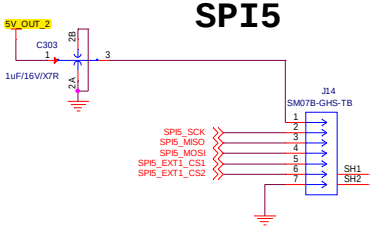
I2C1



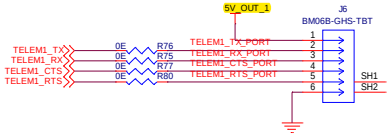
ADC



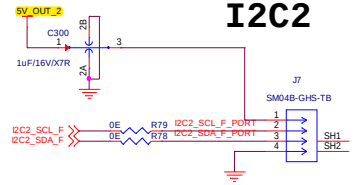
SPI5



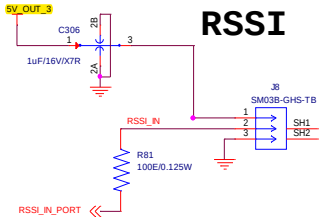
TELEM1



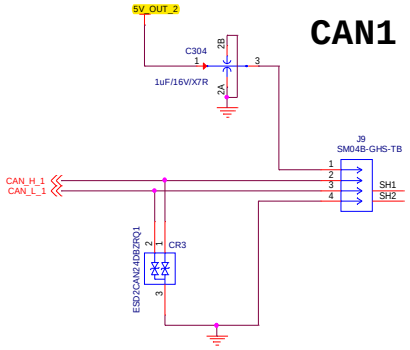
I2C2



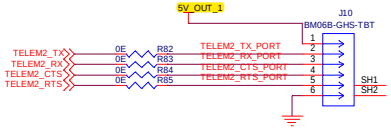
RSSI



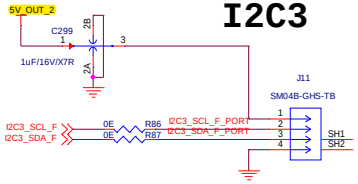
CAN1



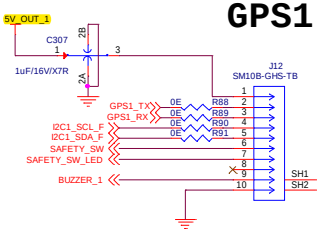
TELEM2



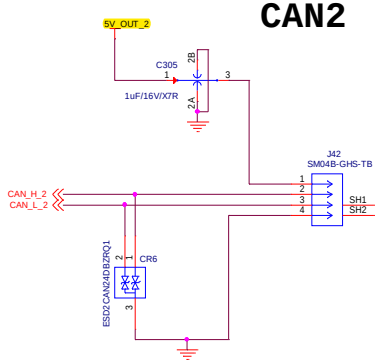
I2C3



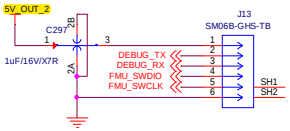
GPS1



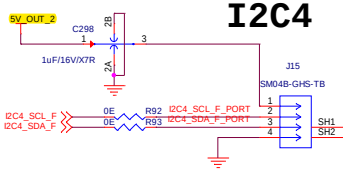
CAN2



DSU7- FMU DEBUG



I2C4

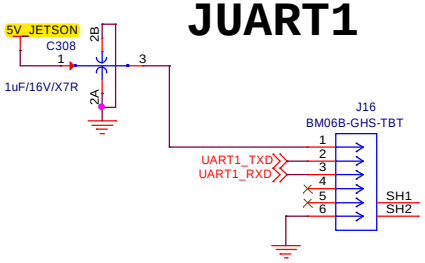


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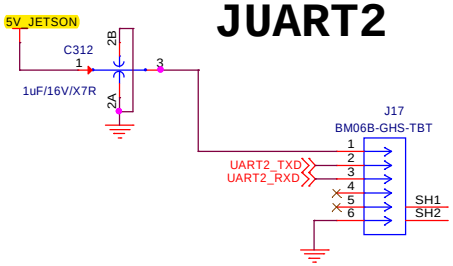
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JETSON I/O Connectors

JUART1

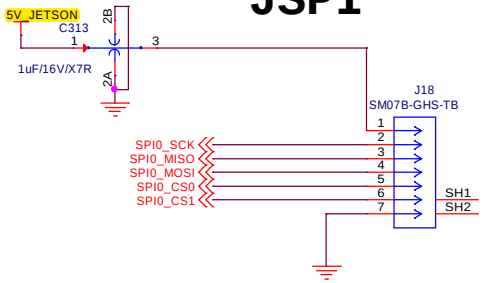


JUART2

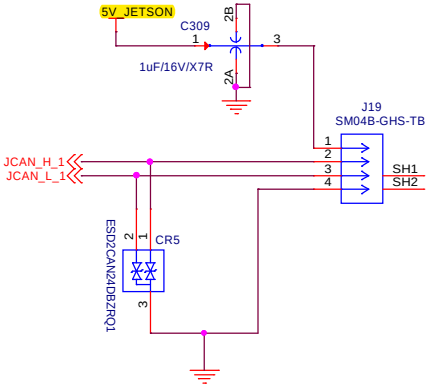


This is the "Debug UART"

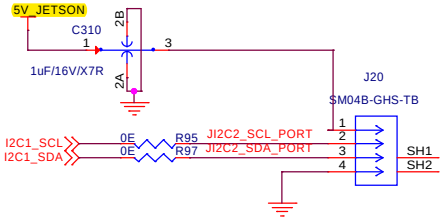
JSPI



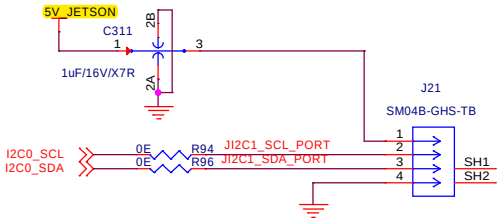
JCAN1



JI2C2

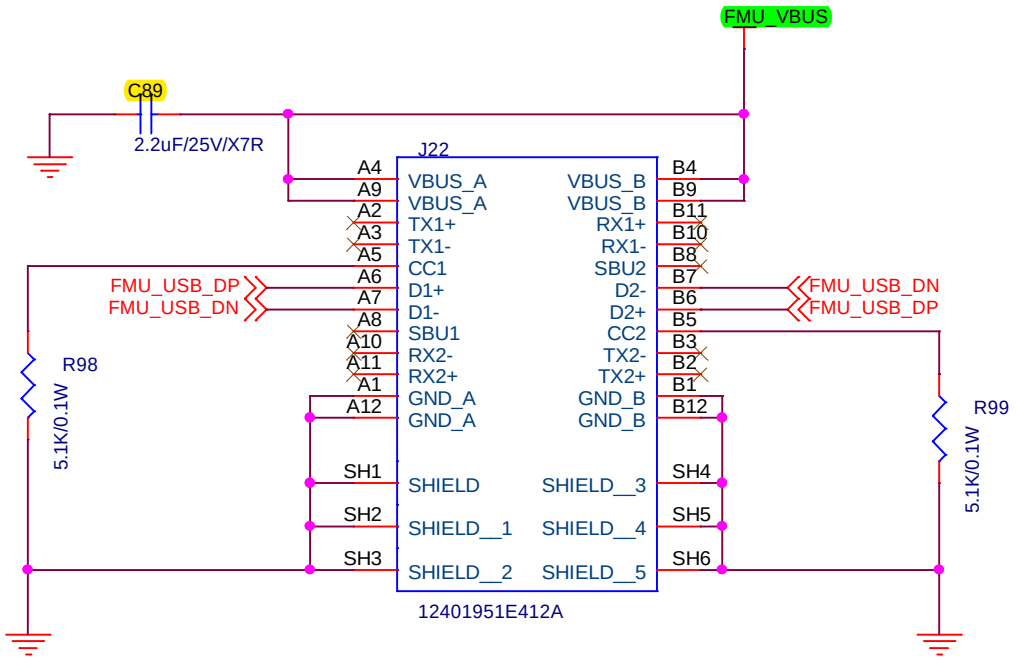


JI2C1

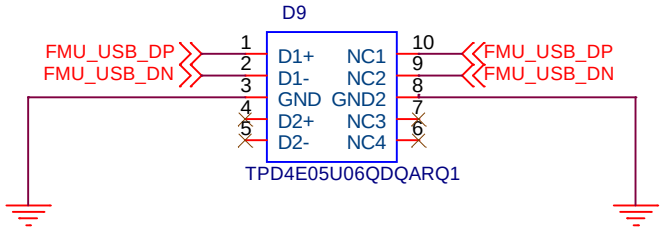


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FCU USB C

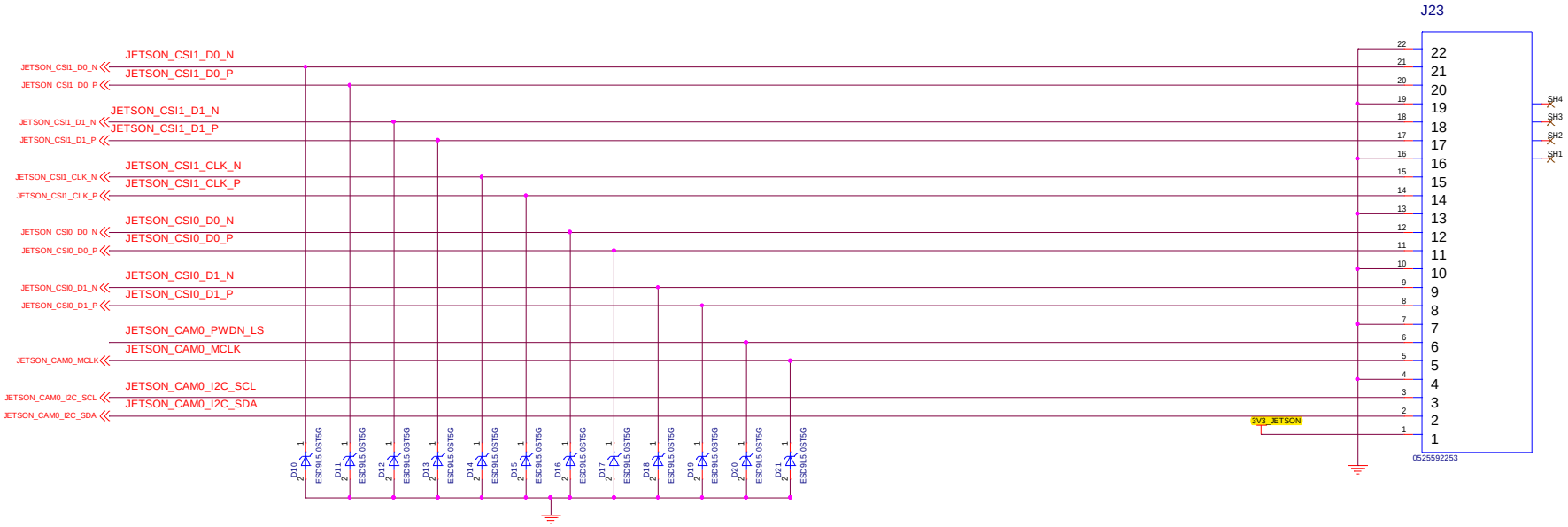


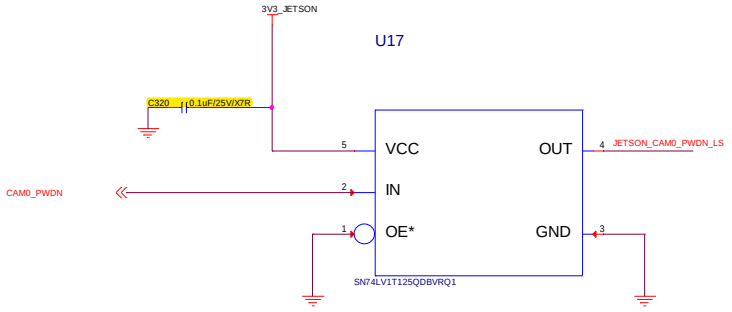
ESD Protection



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CSI CAM0





Note:

x4 camera support in current software only works with the lower clock lane of the x4 clock interface.

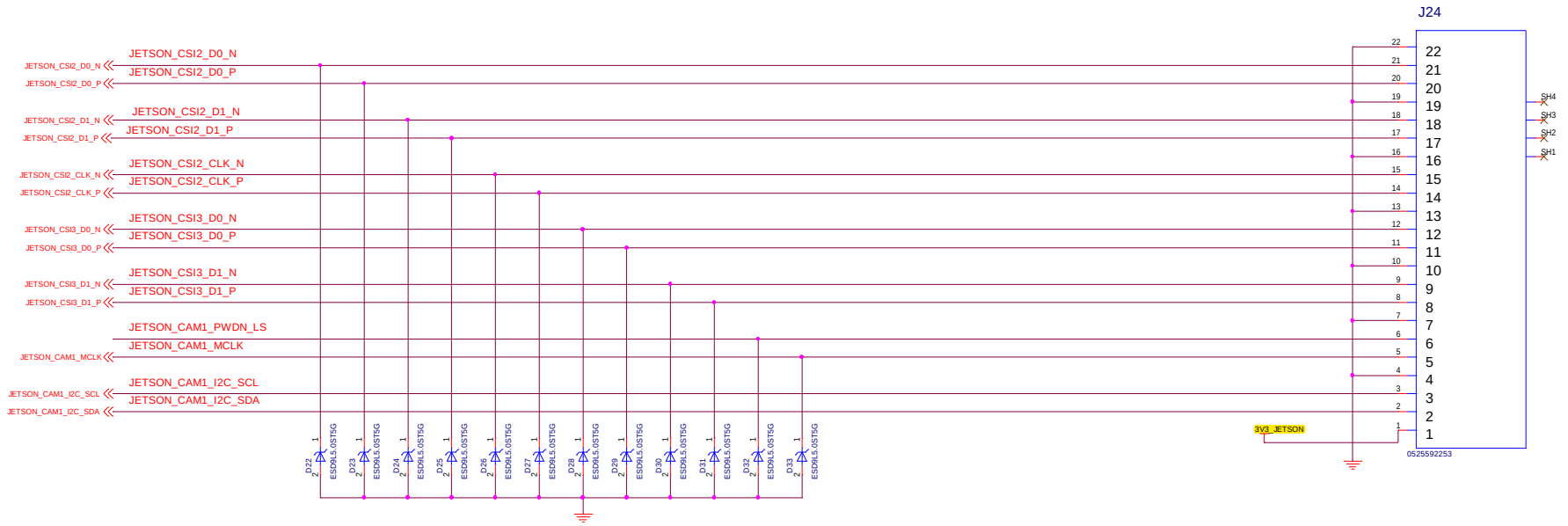
This means that CSI0_CLK or CSI2_CLK are supported for x4 configurations, but not CSI1_CLK or CSI3_CLK.

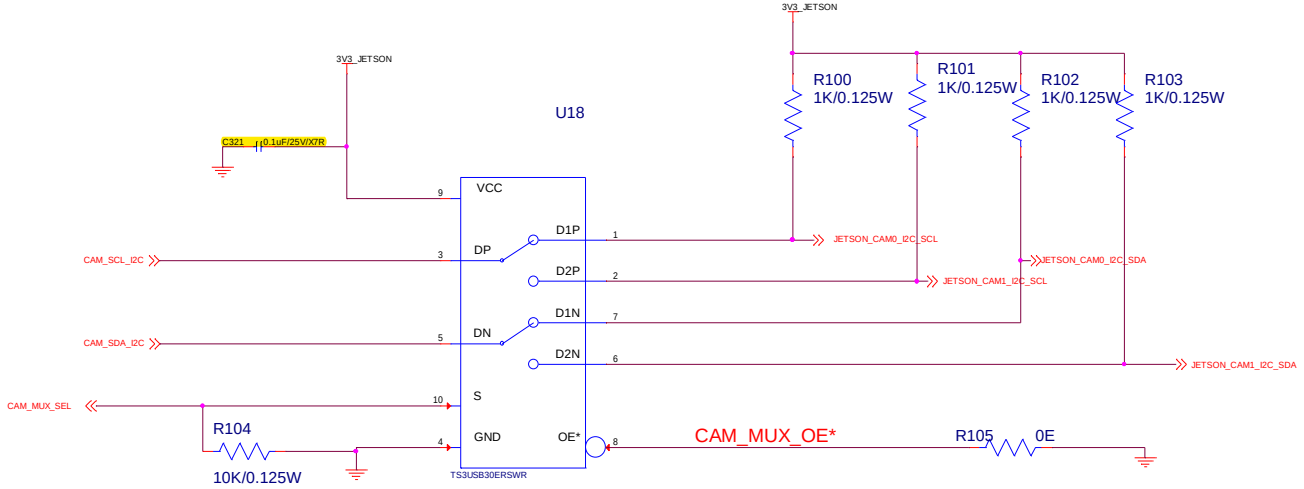
Since J20 has CSI1_CLK, only a x2 camera interface is supported.

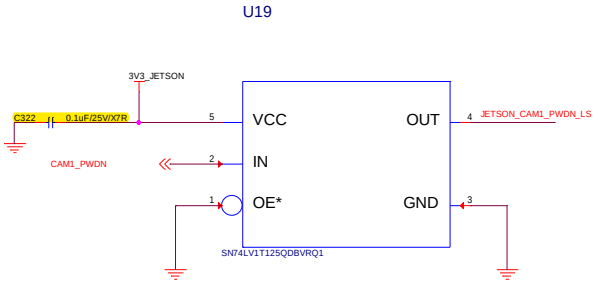
J21 which uses CSI2_CLK can support both x2 or x4 camera interfaces.

DIGITAL SHARK TECHNOLOGY		
Title: CSI CAM 0		
Size: C	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
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CSI CAM1



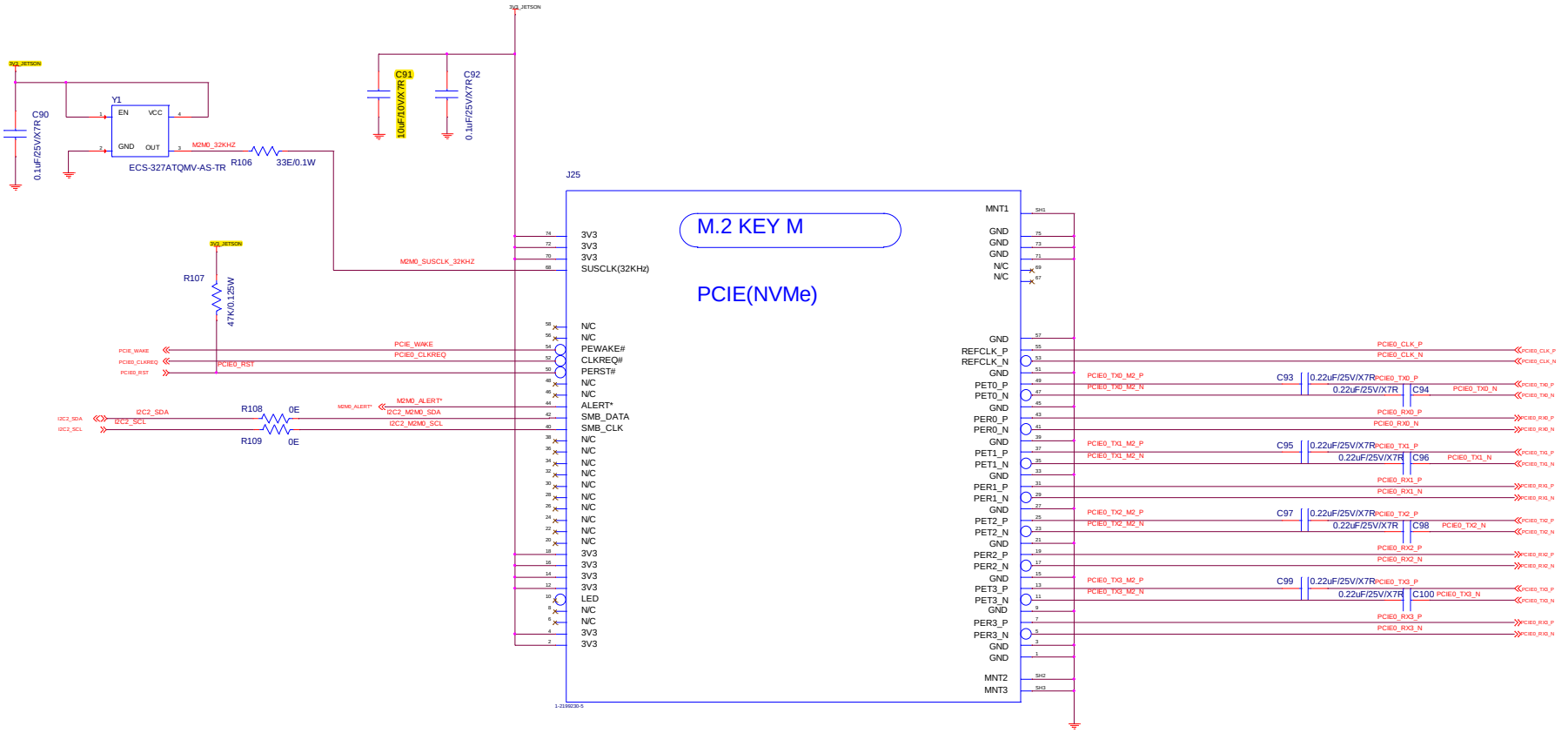




DIGITAL SHARK TECHNOLOGY		
Title: CSI CAM 1		
Size: C	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
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M.2 Key M1 - 2230

M.2 Key M1 (2230) Ÿ Compact NVMe SSD for OS boot.



DIGITAL SHARK TECHNOLOGY

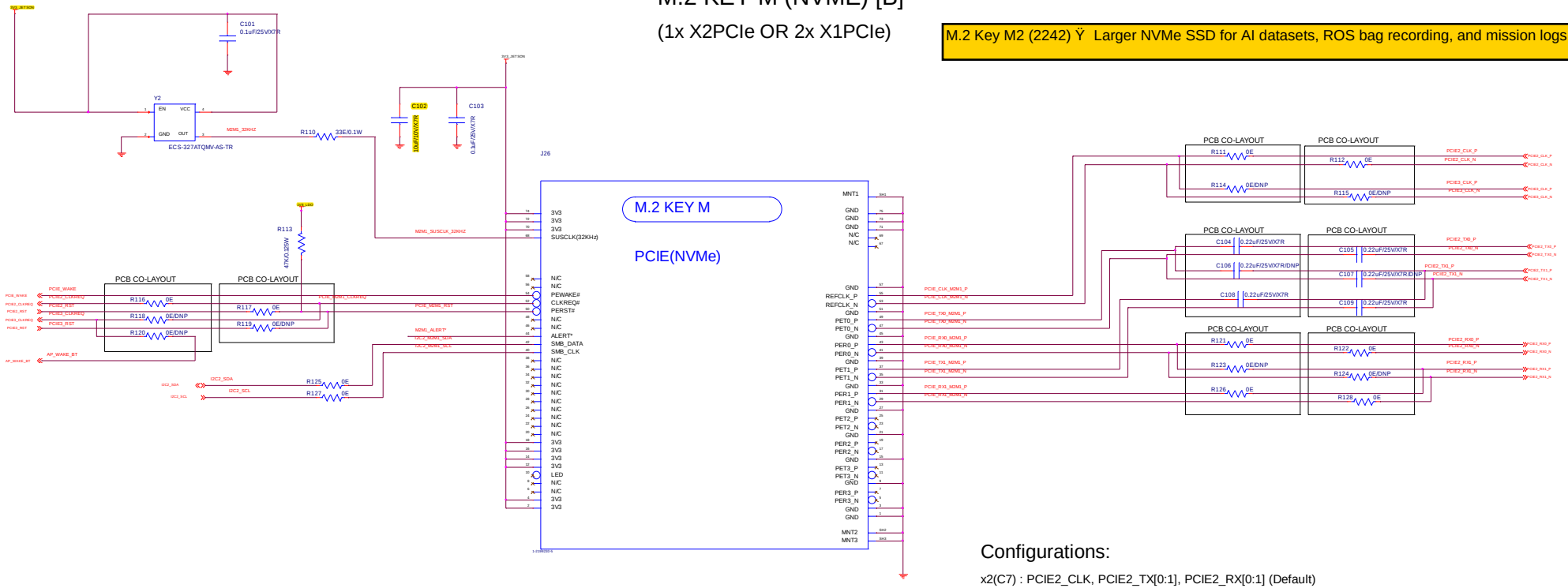
Title	M.2 Key M1	
Size: D	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
Date: Wednesday, December 03, 2025	Sheet 17 of 34	

M.2 Key M2 - 2242

M.2 KEY-M (NVME) [B]

(1x X2PCIe OR 2x X1PCIe)

M.2 Key M2 (2242) Ÿ Larger NVMe SSD for AI datasets, ROS bag recording, and mission logs.

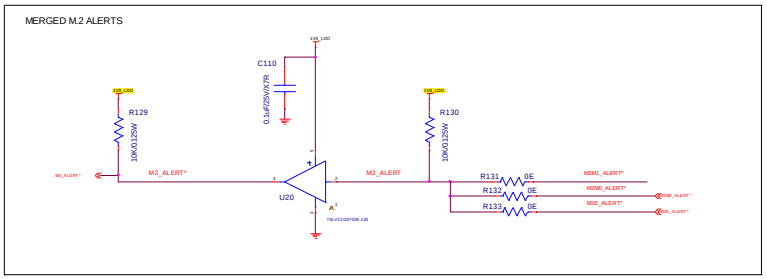


Configurations:

x2(C7) : PCIE2_CLK, PCIE2_TX[0:1], PCIE2_RX[0:1] (Default)

x1(C7) : PCIE2_CLK, PCIE2_TX0, PCIE2_RX0

x1(C9) : PCIE3_CLK, PCIE2_TX1, PCIE2_RX1



DIGITAL SHARK TECHNOLOGY

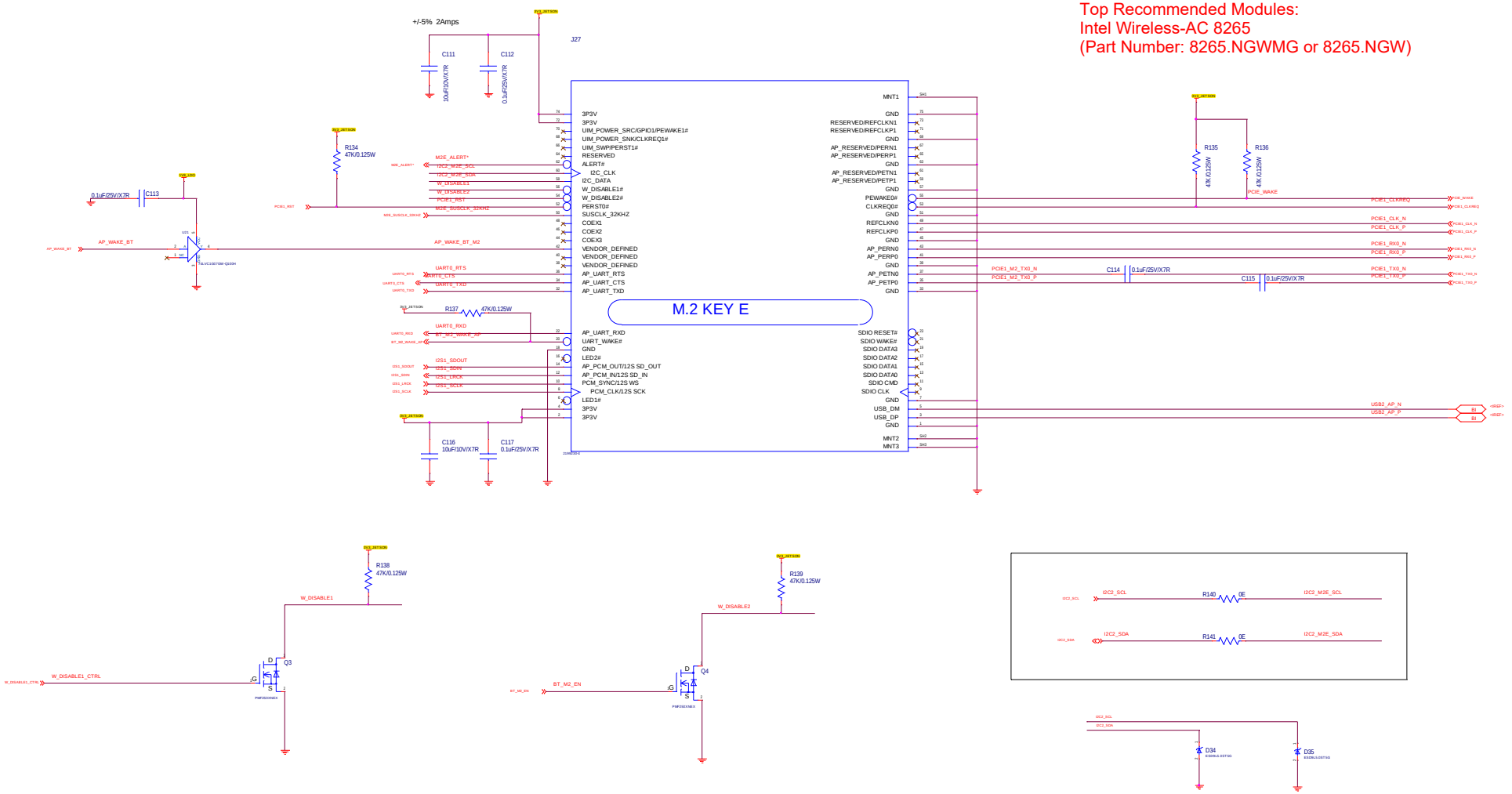
M.2 Key M2

JETSON BASEBOARD V1.1

M.2 Key E - 2230

M.2 Key E (2230) → Wi-Fi 6/6E/7 + Bluetooth module.

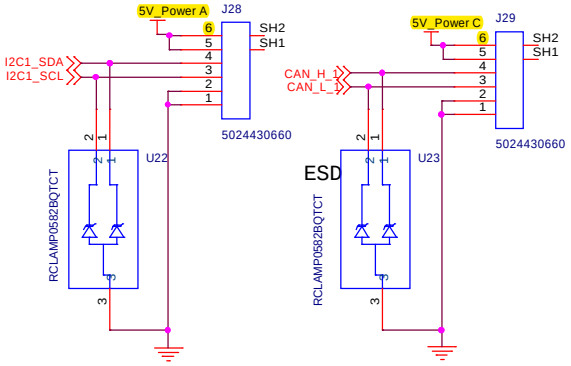
Top Recommended Modules:
Intel Wireless-AC 8265
(Part Number: 8265.NGWMG or 8265.NGW)



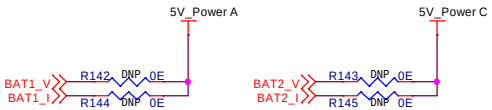
DIGITAL SHARK TECHNOLOGY	
Title	M.2 Key E
Size	Document Number: JETSON BASEBOARD V1.1
Date	Rev: 1.1

POWER INPUTS

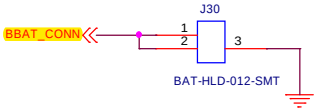
Power Module Connectors



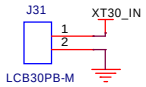
Battery V & I Sensing



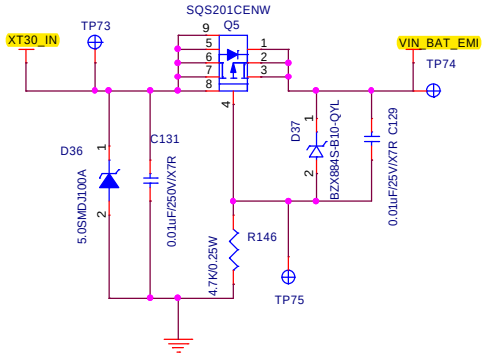
RTC Battery Connector



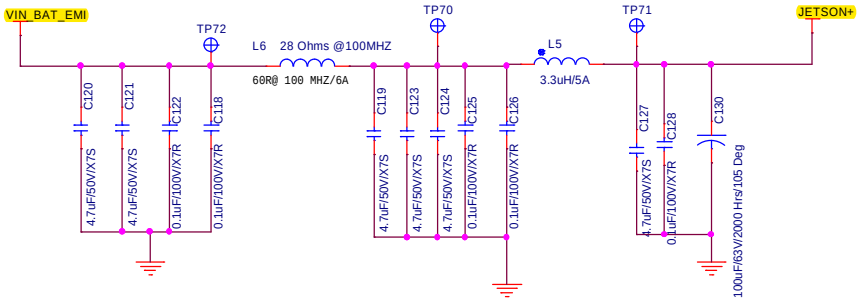
XT60 Connector



PROTECTION CIRCUIT

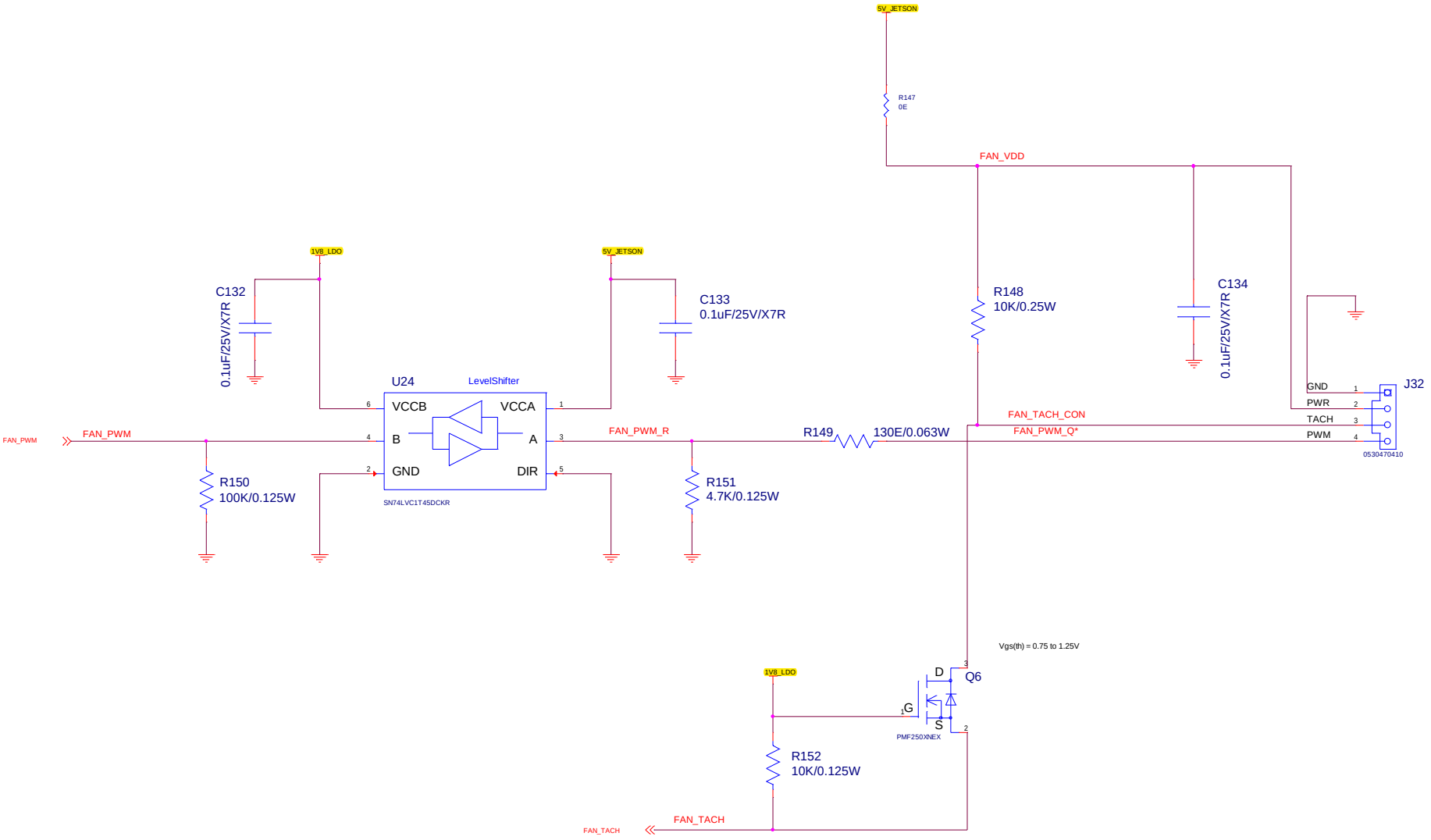


EMI FILTER CIRCUIT



DIGITAL SHARK TECHNOLOGY		
Title: Power Inputs		
Size: B	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
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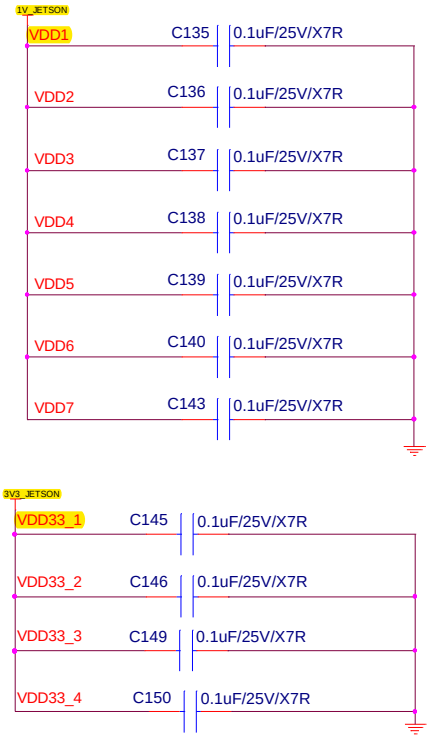
PWM FAN Connector

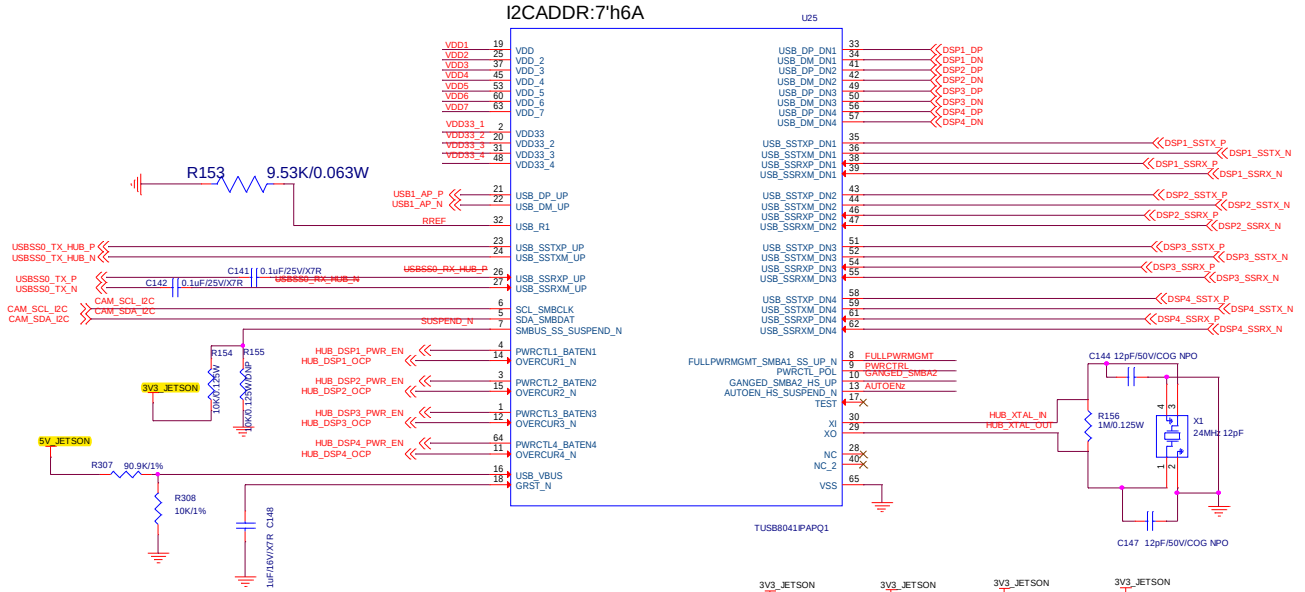


DIGITAL SHARK TECHNOLOGY		
Title: JETSON PWM Fan		
Size: C	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
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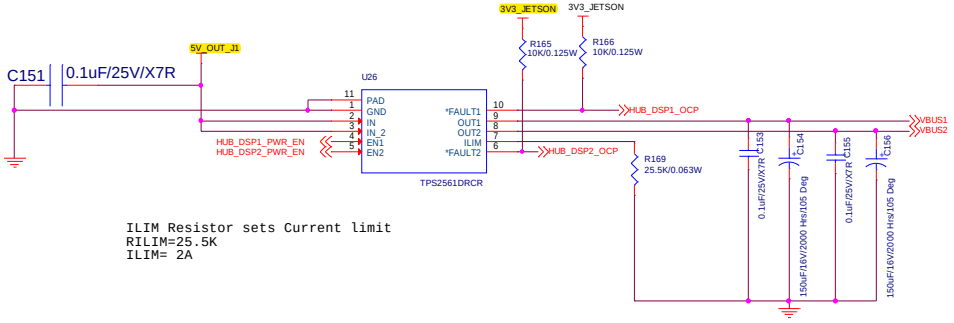
USB HUB Controller

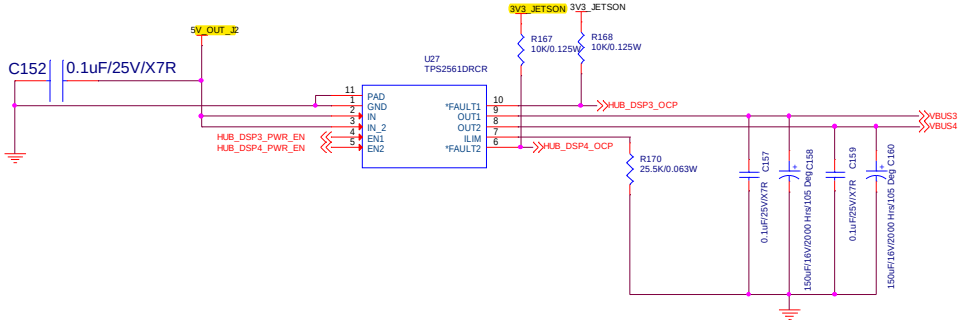
DECOUPLING CAPS





USB HUB Current Limiter

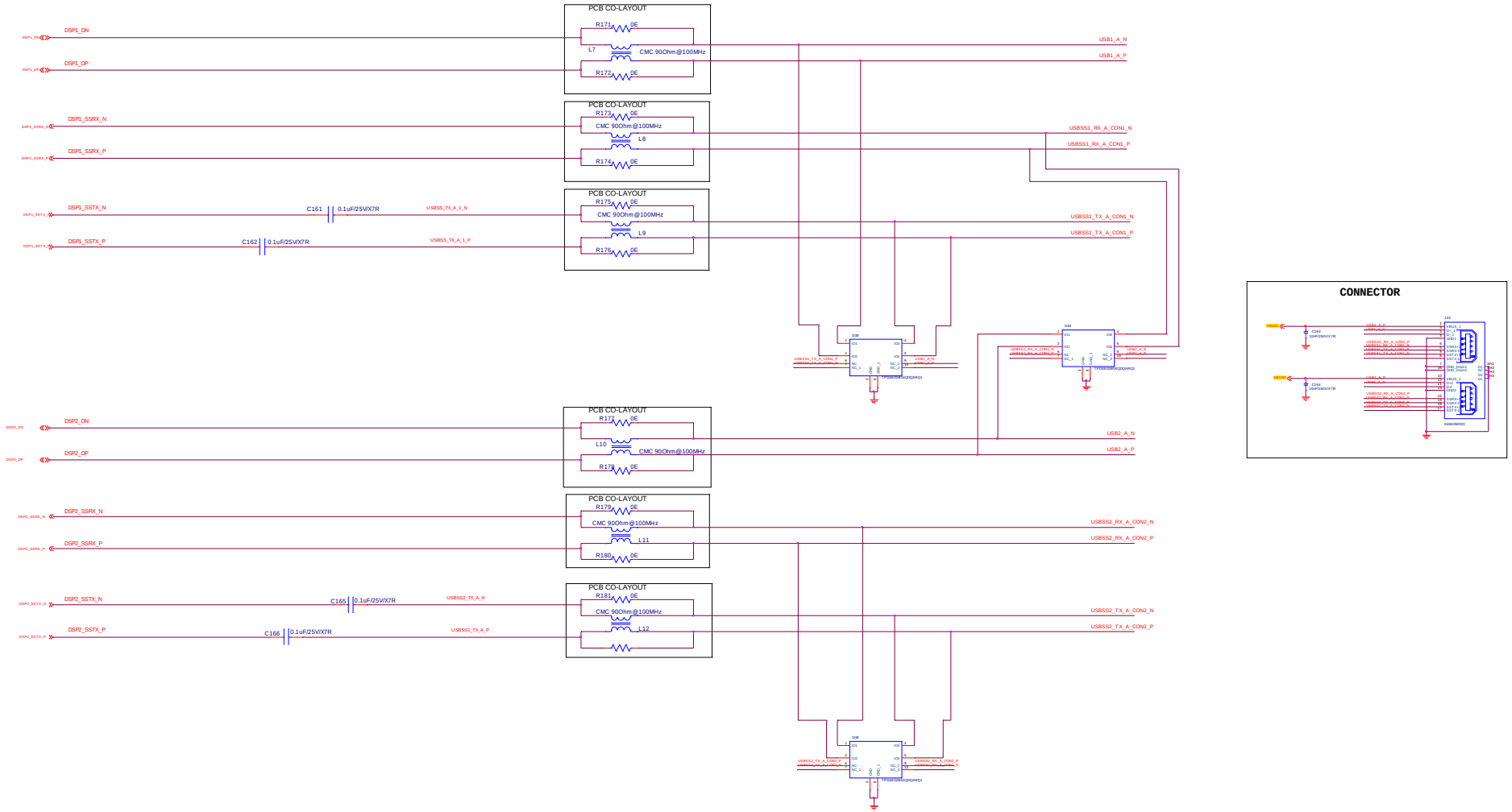




ILIM Resistor sets Current limit
RILIM=25.5K
ILIM= 2A

DIGITAL SHARK TECHNOLOGY		
Title: USB Hub Controller		
Size: C	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
Date: Wednesday, December 03, 2025	Sheet 22 of 34	

USB 3.1 Type A x 2 [A]

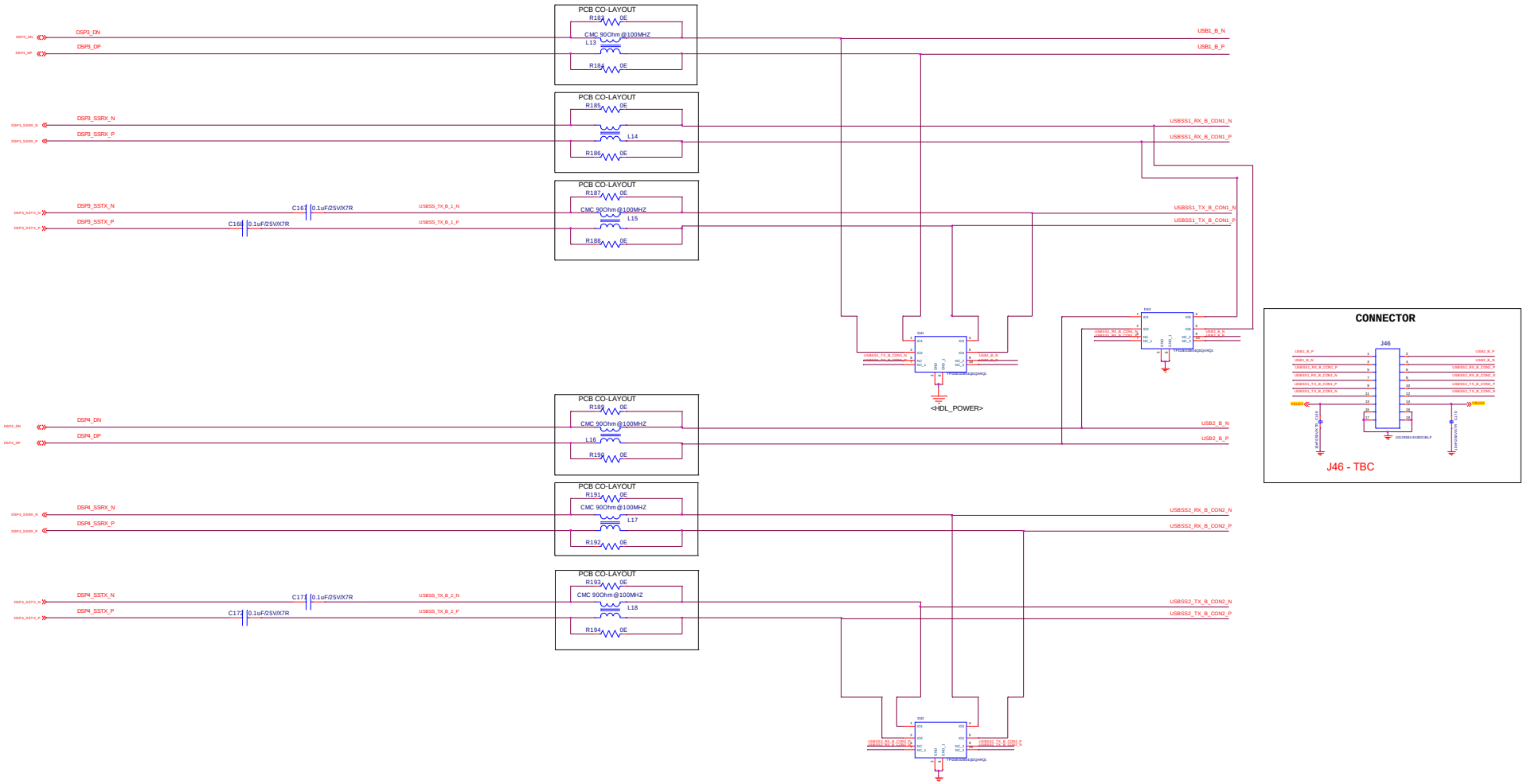


DIGITAL SHARK TECHNOLOGY

JETSON USB A 1

JETSON BASEBOARD V1.3

USB 3.1 Type A x 2 [B]

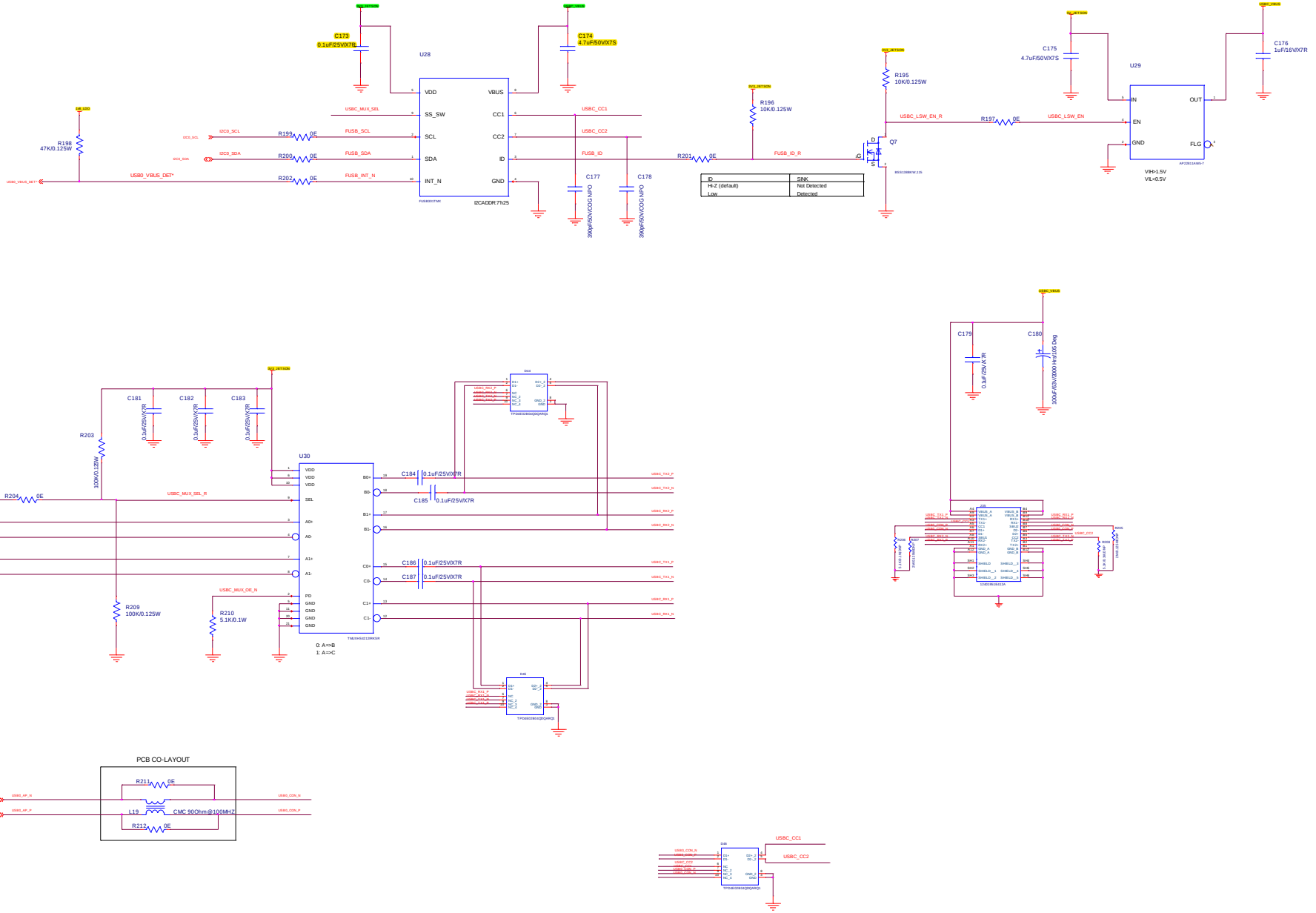


DIGITAL SHARK TECHNOLOGY

JETSON USB A 3

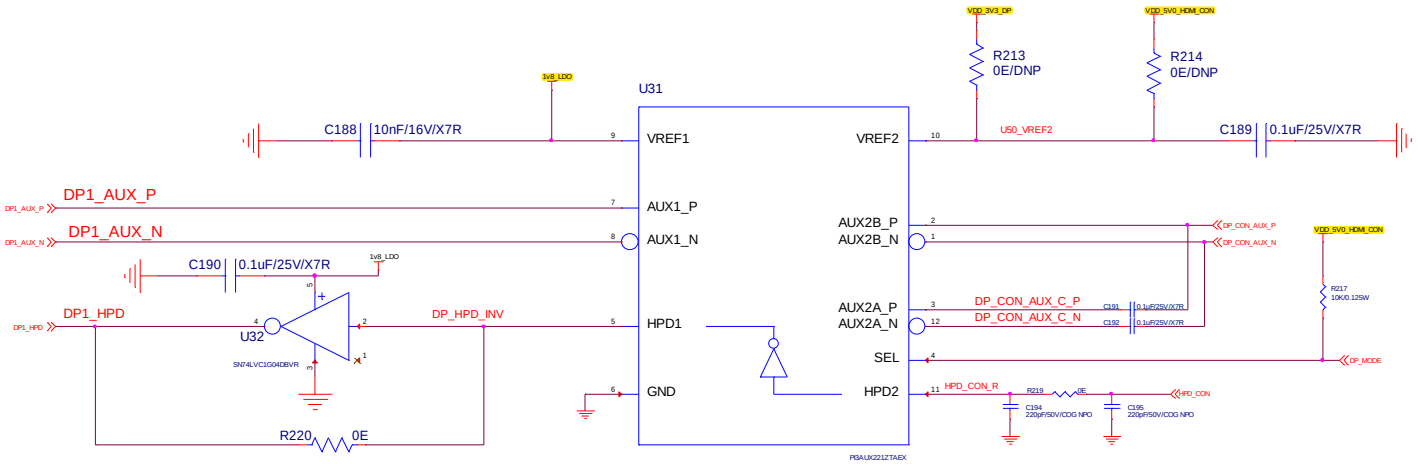
JETSON BASEBOARD V1.3

JETSON USB C

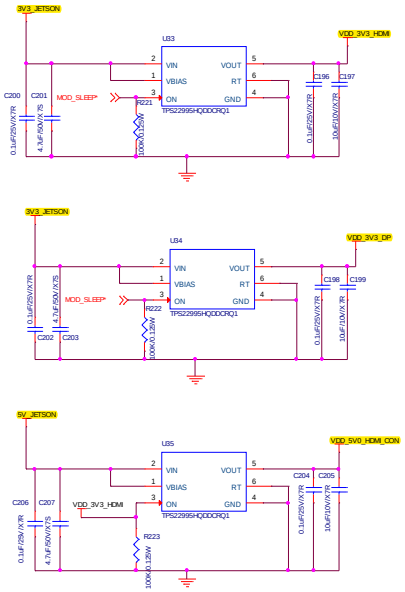


DIGITAL SHARK TECHNOLOGY	
JETSON USBC	
DOCUMENT NUMBER: JETSON BASEBOARD V1.1	Rev: 1.1

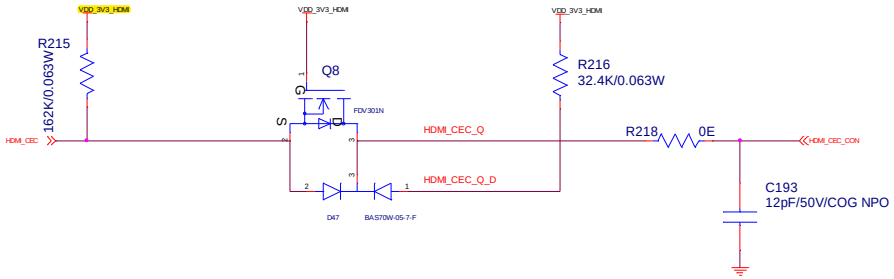
HDMI/DP MUX



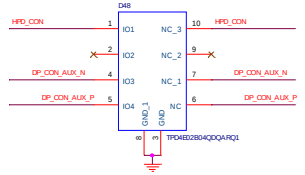
Port Load Switches



CEC Circuit



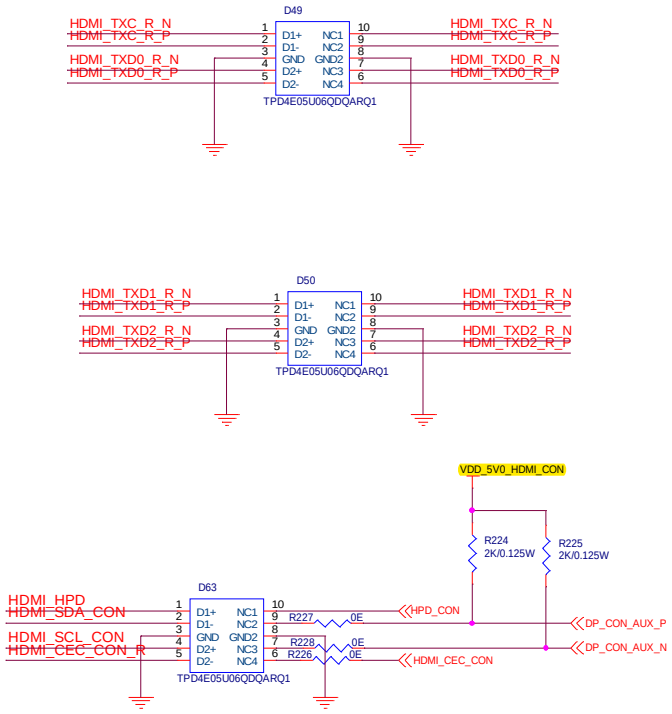
ESD Protection



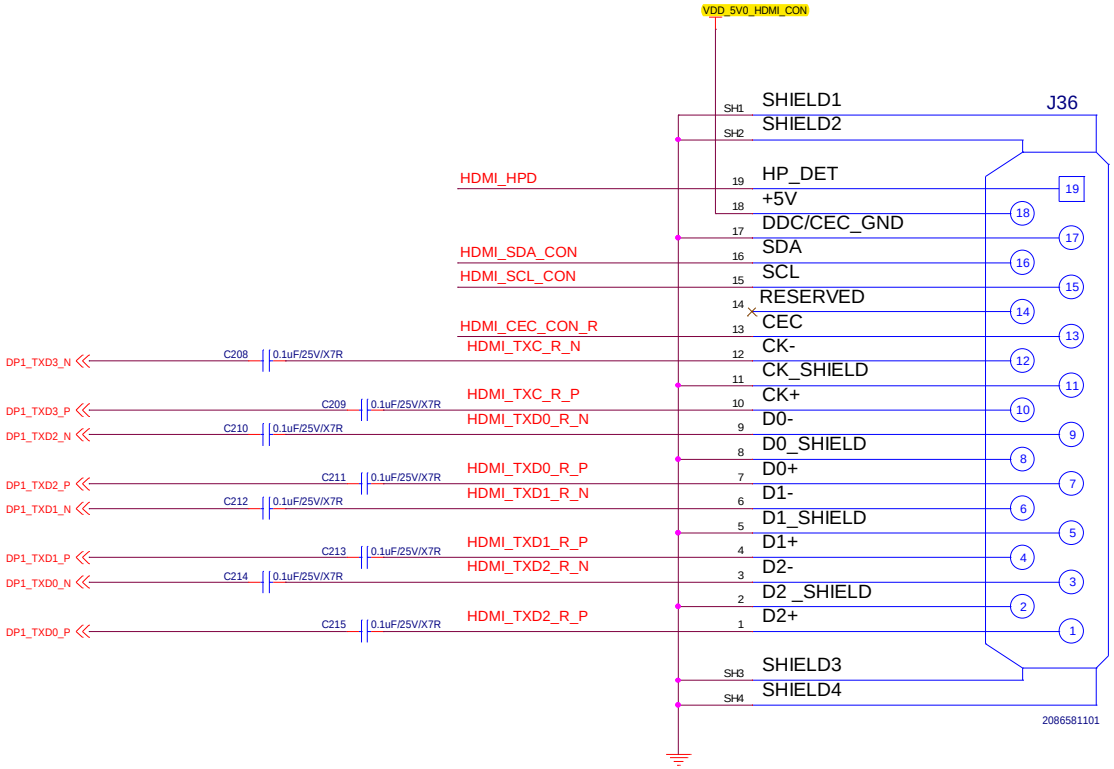
DIGITAL SHARK TECHNOLOGY		
Title: JETSON HDMI/DP		
Size: Custom	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
Date: Wednesday, December 03, 2025	Sheet: 26 of 34	

HDMI Port

ESD Protection



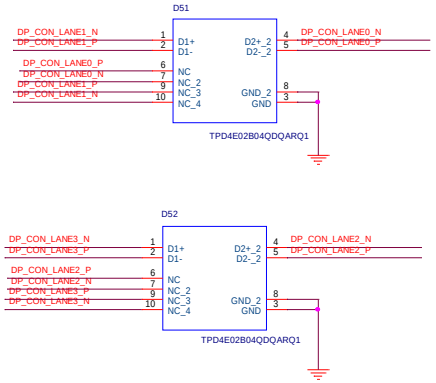
CONNECTOR



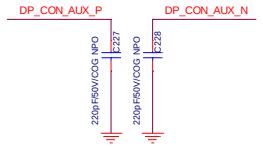
DIGITAL SHARK TECHNOLOGY		
Title: HDMI PORT		
Size: Custom	Document Number: JETSON BASEBOARD V1.1	Rev: 1.0
Date: Wednesday, December 03, 2025	Sheet 27 of 34	

Display Port

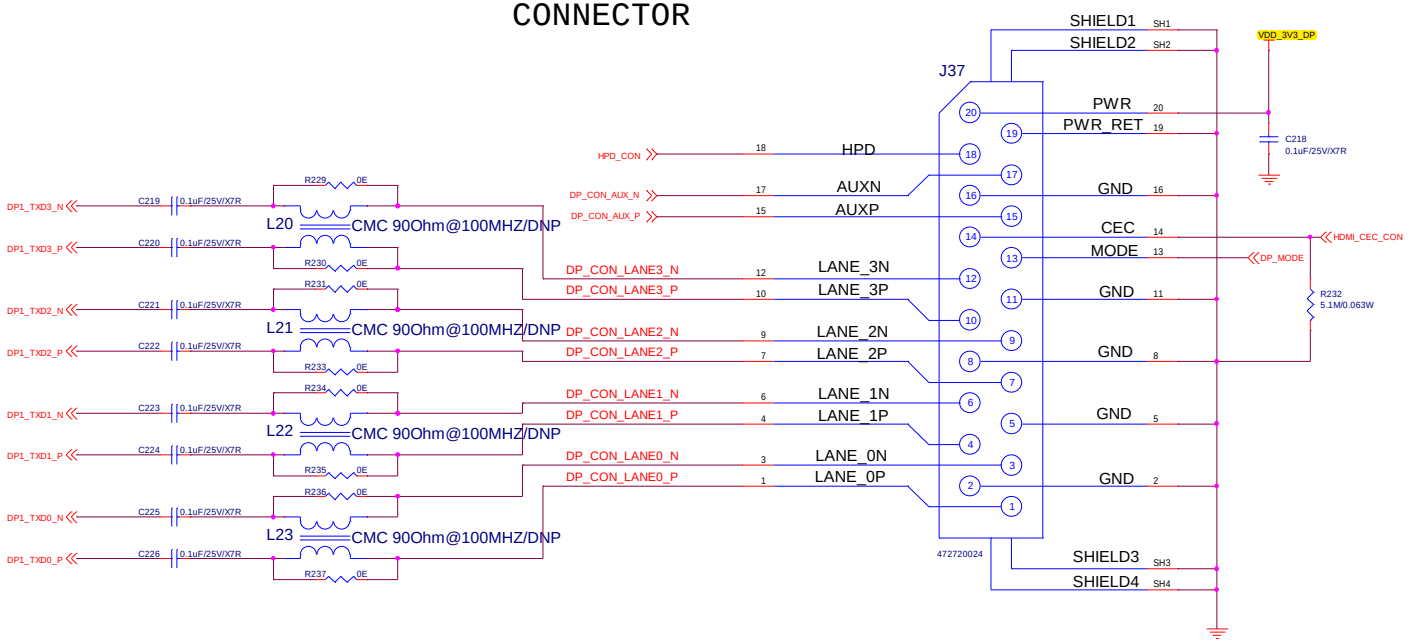
ESD Protection



De-Coupling Capacitors

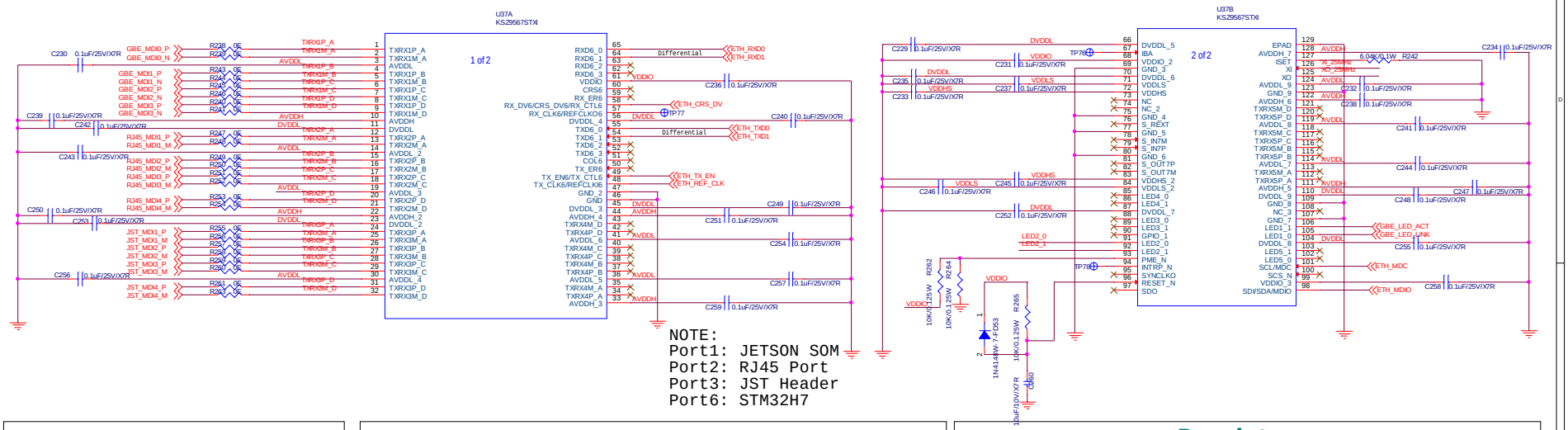


CONNECTOR

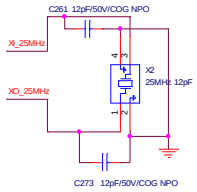


DIGITAL SHARK TECHNOLOGY			
Title: Display Port			
Size: Custom	Document Number: JETSON BASEBOARD V1.1		Rev: 1.0
Date: Wednesday, December 03, 2025	Sheet 28 of 34		

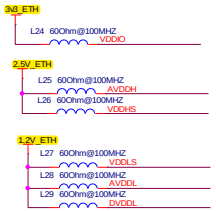
ETHERNET SWITCH



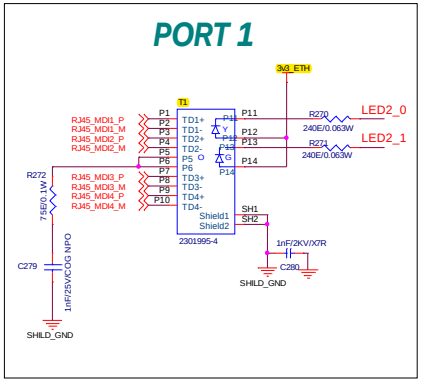
25MHz Crystal



Power Connections

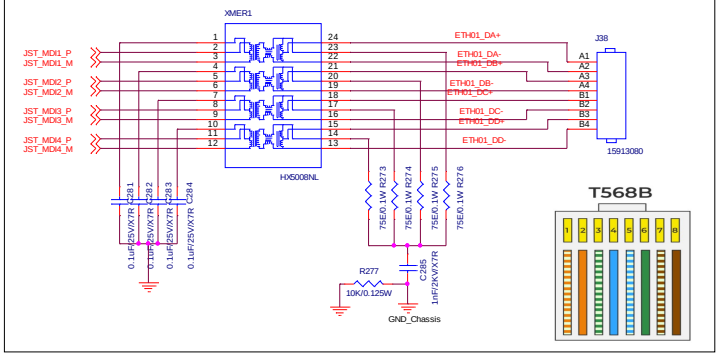


PORT 1

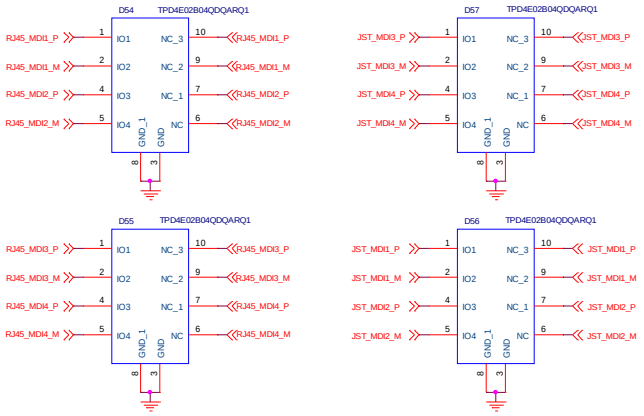


PORT 3

MAGNETICS JST HEADER

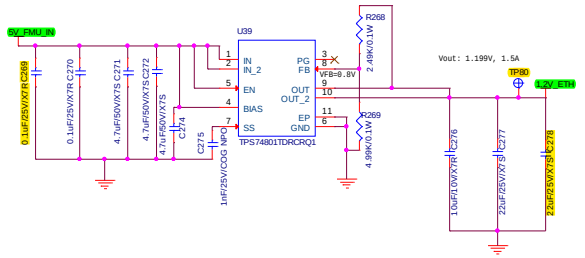


ESD Protection Diode Array

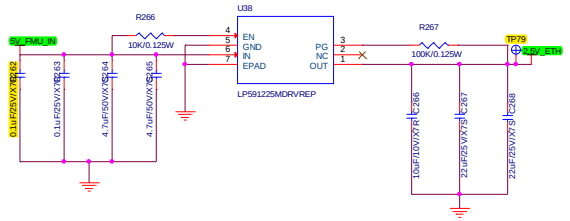


Regulators

1.2V 1.5A

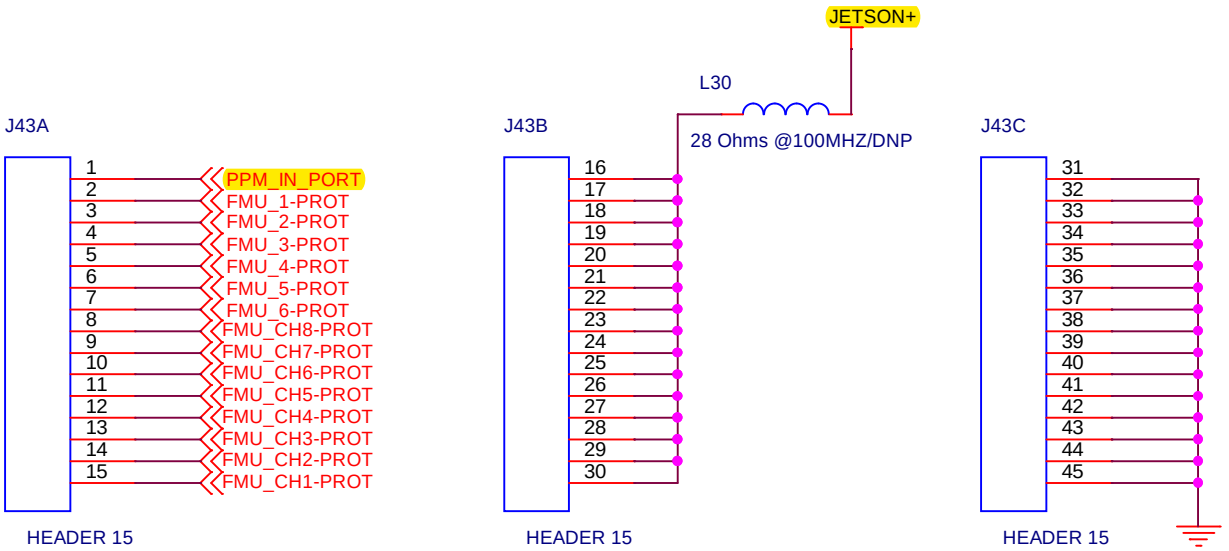


2.5V 500mA



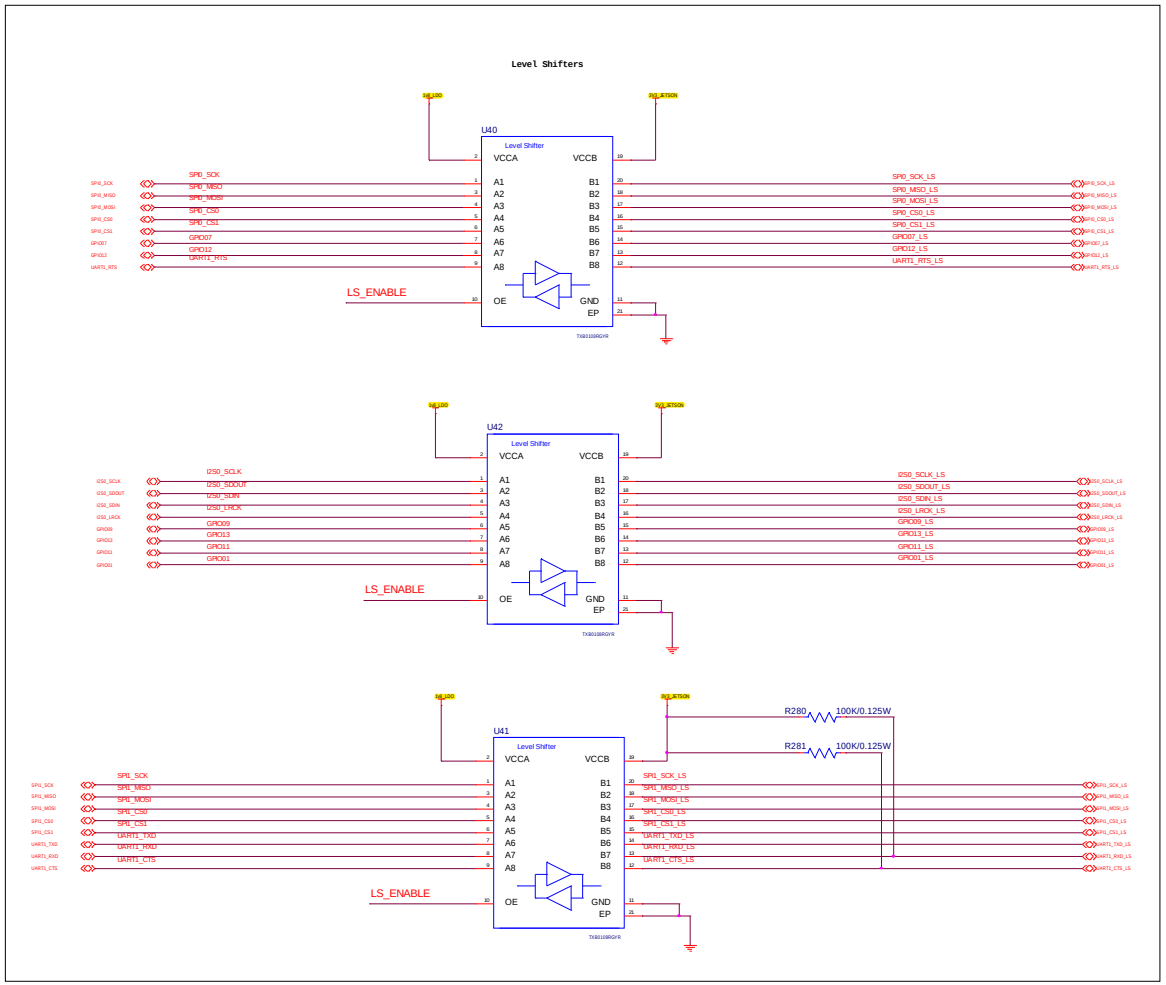
DIGITAL SHARK TECHNOLOGY			
Title: Ethernet Switch			
Size: Custom	Document Number: JETSON BASEBOARD V1.1		Rev: 1.1
Date: Wednesday, December 03, 2025	Sheet: 29	of: 34	

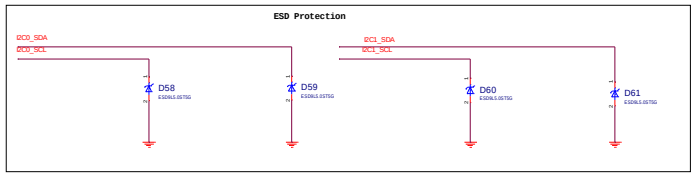
PWM Output

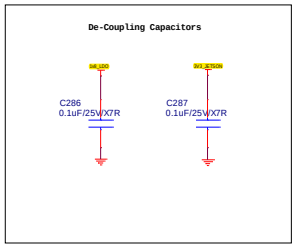


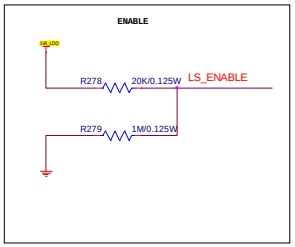
DIGITAL SHARK TECHNOLOGY		
Title: PWM Output		
Size: A	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
Date: Wednesday, December 03, 2025	Sheet 30 of 34	

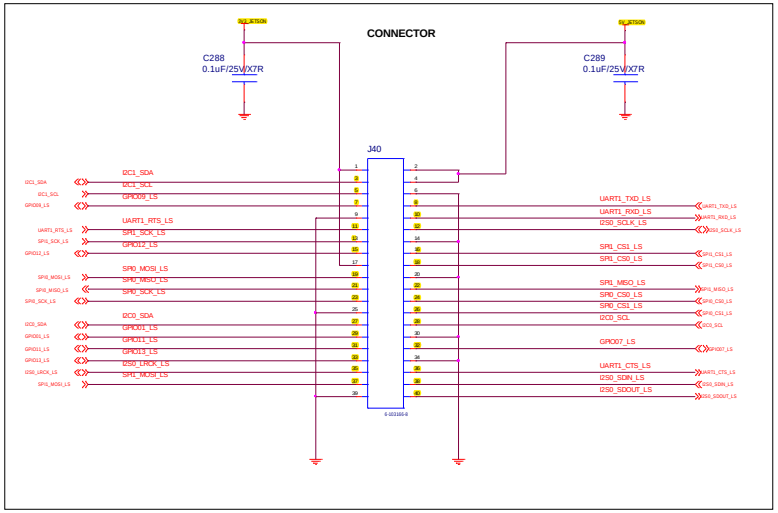
40 Pin GPIO Expander











DIGITAL SHARK TECHNOLOGY

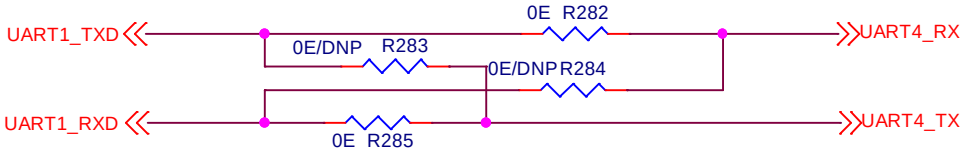
Title	GPIO Header		
Size: Custom	Document Number: JETSON BASEBOARD V1.1		Rev: 1.1
Date:	Wednesday, December 09, 2015	Sheet: 16 of 36	

Board-to-Board Interface

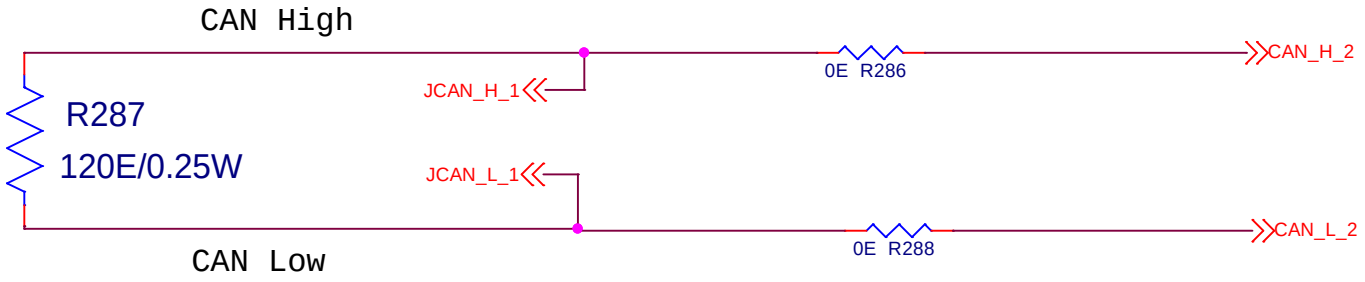
JETSON

FCU

UART

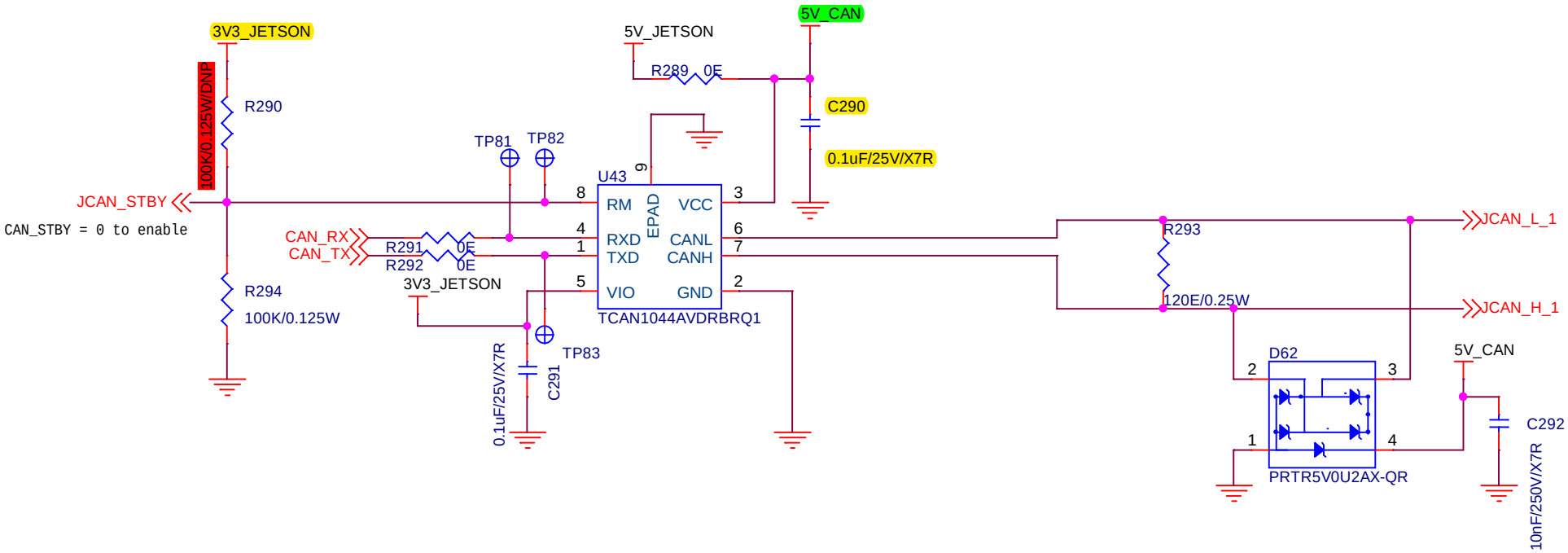


CAN BUS



DIGITAL SHARK TECHNOLOGY		
Title: Board-to-Board Interface		
Size: A	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
Date: Wednesday, December 03, 2025	Sheet 32 of 34	

CAN Bus - Jetson

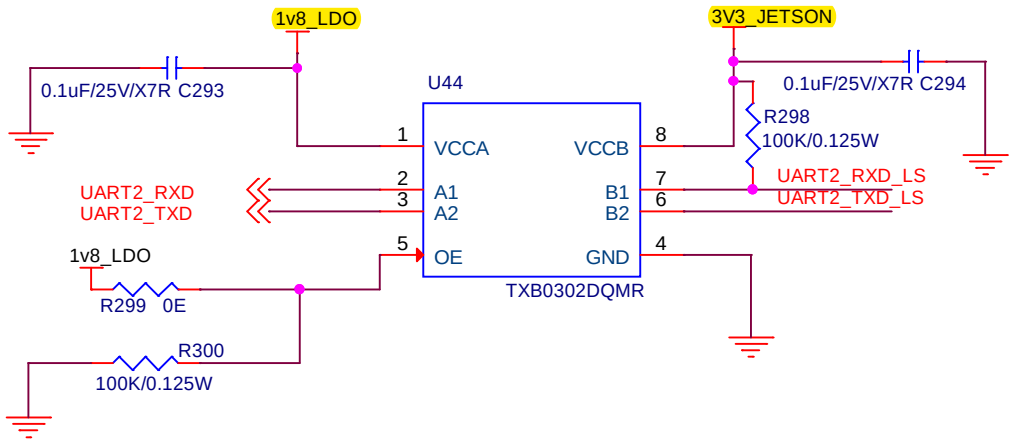


DIGITAL SHARK TECHNOLOGY

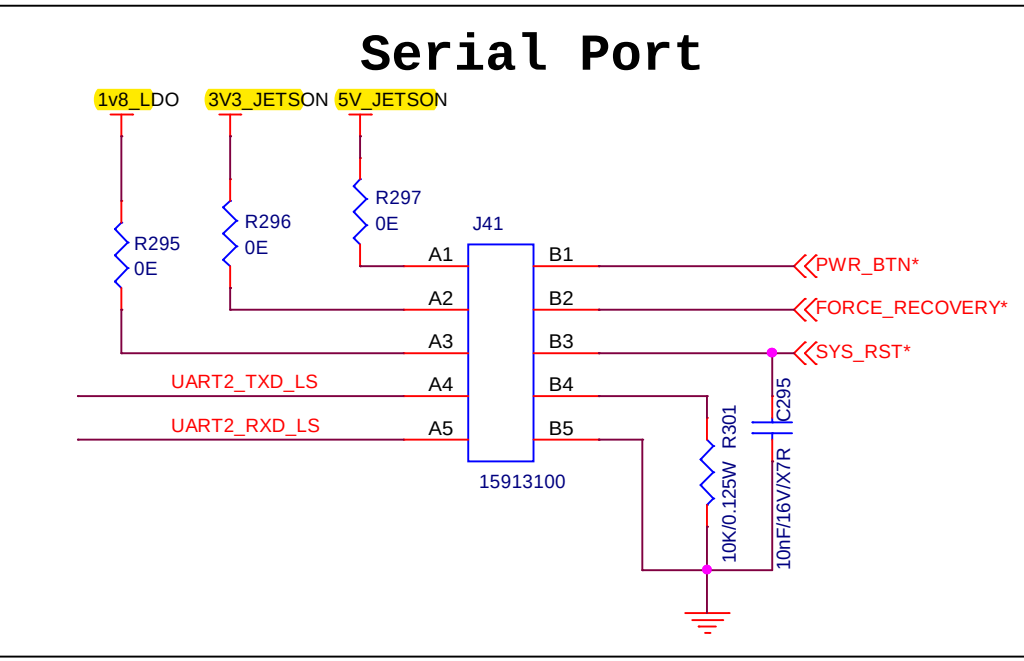
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Size: A	Document Number: JETSON BASEBOARD V1.1	Rev: 1.1
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Serial Header

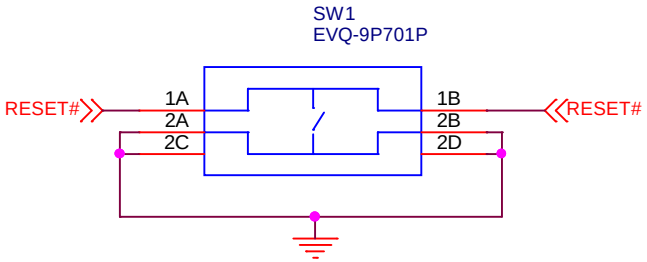
Serial Port



Serial Port



STM Reset Button



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Title: Serial Header		
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